NEURORADIOLOGY

Reading Brain Imaging

One 73-year-old patient, five questions asked in the same order every time, and the sequences that answer them. A complete walk through Dr. Rybinnik's crash course, from landmarks on a plain head CT to the enhancement pattern that finally names the lesion.

CONTENTS

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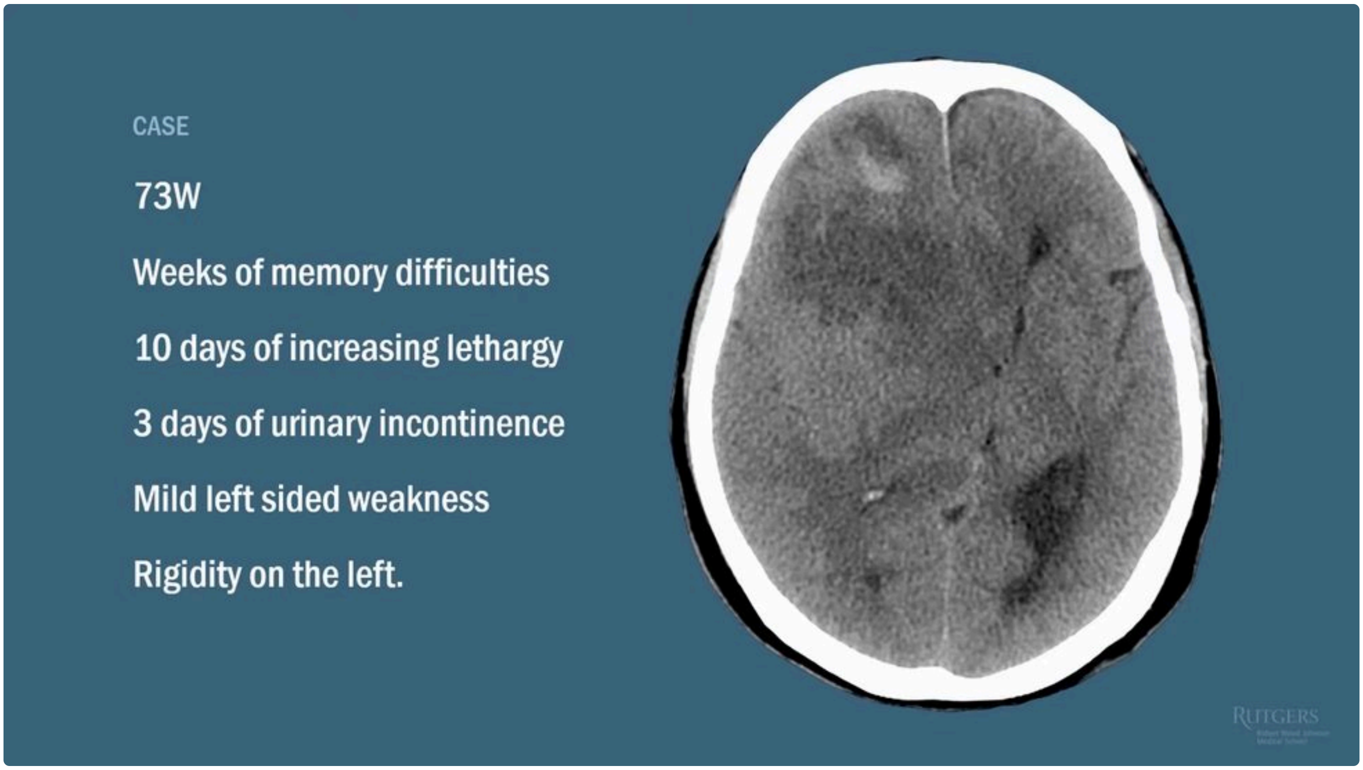
01 The Case

One patient runs through this whole guide. Everything else is the machinery you need to solve her.



The presenting scan. Head CT was already done by the ED team.

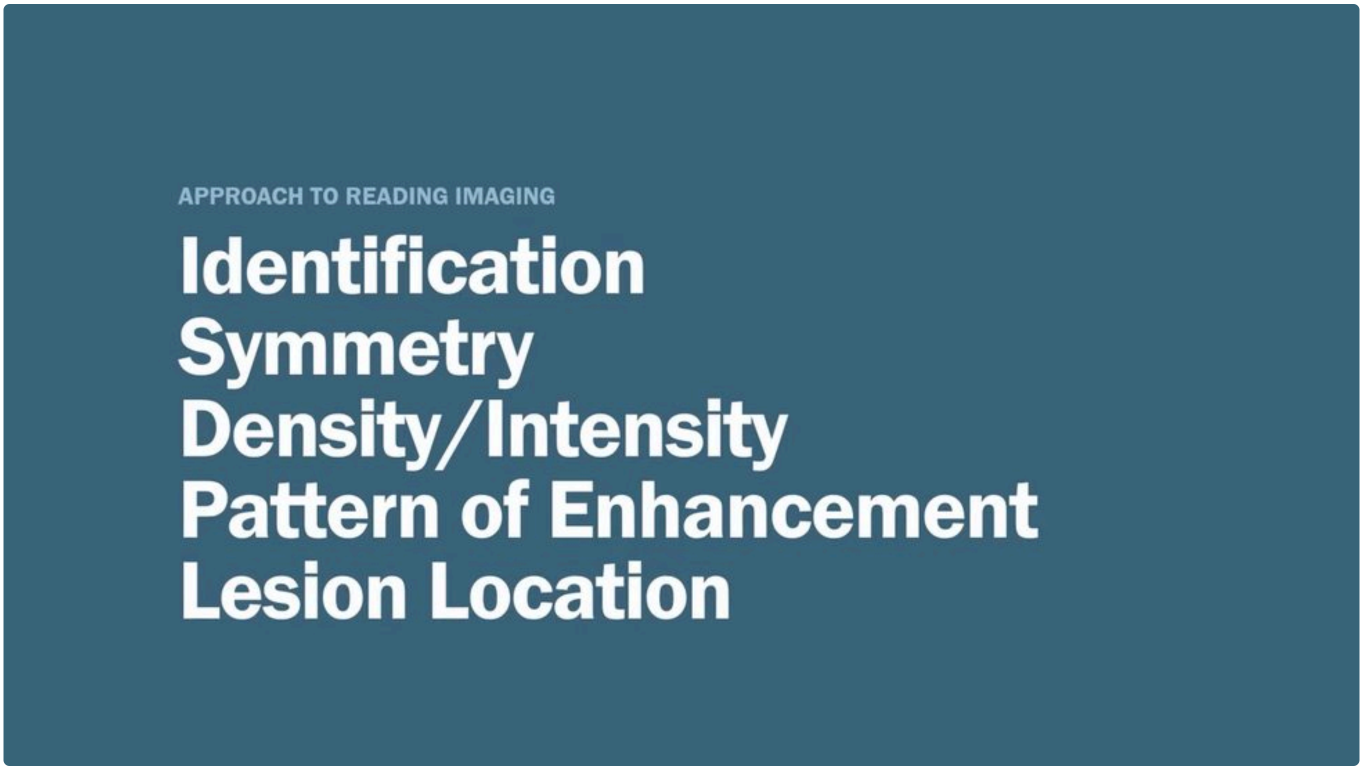
- **73-year-old woman, previously healthy.** Weeks of memory difficulty, 10 days of increasing lethargy, 3 days of urinary incontinence, and now mild left-sided weakness with rigidity.
- **You are alone with it.** The radiology resident has passed out and the neurology resident is running a stroke code.

THE RULE THAT OUTRANKS EVERY IMAGE

It is the **clinical history**, not the imaging, that gives you the diagnosis. Imaging at its best gives you a reasonable differential. Read the history first, every time.

02 The Approach

When you face an unknown scan you need an order of operations. This is it, and the rest of the guide follows it exactly.



The five questions, in order. Work them top to bottom.

#	STEP	WHAT YOU ARE ASKING
1	Identification	Which scan, which sequence, which slice?
2	Symmetry	Is anything asymmetric, and where is midline?
3	Density / intensity	Is the lesion bright or dark, and on which sequence?
4	Enhancement	Does it take contrast, and in what pattern?
5	Location	Extra-axial (outside the brain) or intra-axial (in the parenchyma)?

Put all five answers in a blender, and what pours out is your differential.

03 Reading A Head CT

The non-contrast head CT is the most common study you will ever be handed, and it is the only CT sequence this guide covers.

STUDY	WHAT IT IS FOR
CT without contrast	The workhorse. Acute blood, bone, gross structure.
CT with contrast	Blood brain barrier breakdown. MRI does this better.
CT angiogram	Vessels. Same contrast, different timing. Powered for vessels, not parenchyma.
CT perfusion	Estimates brain blood flow.

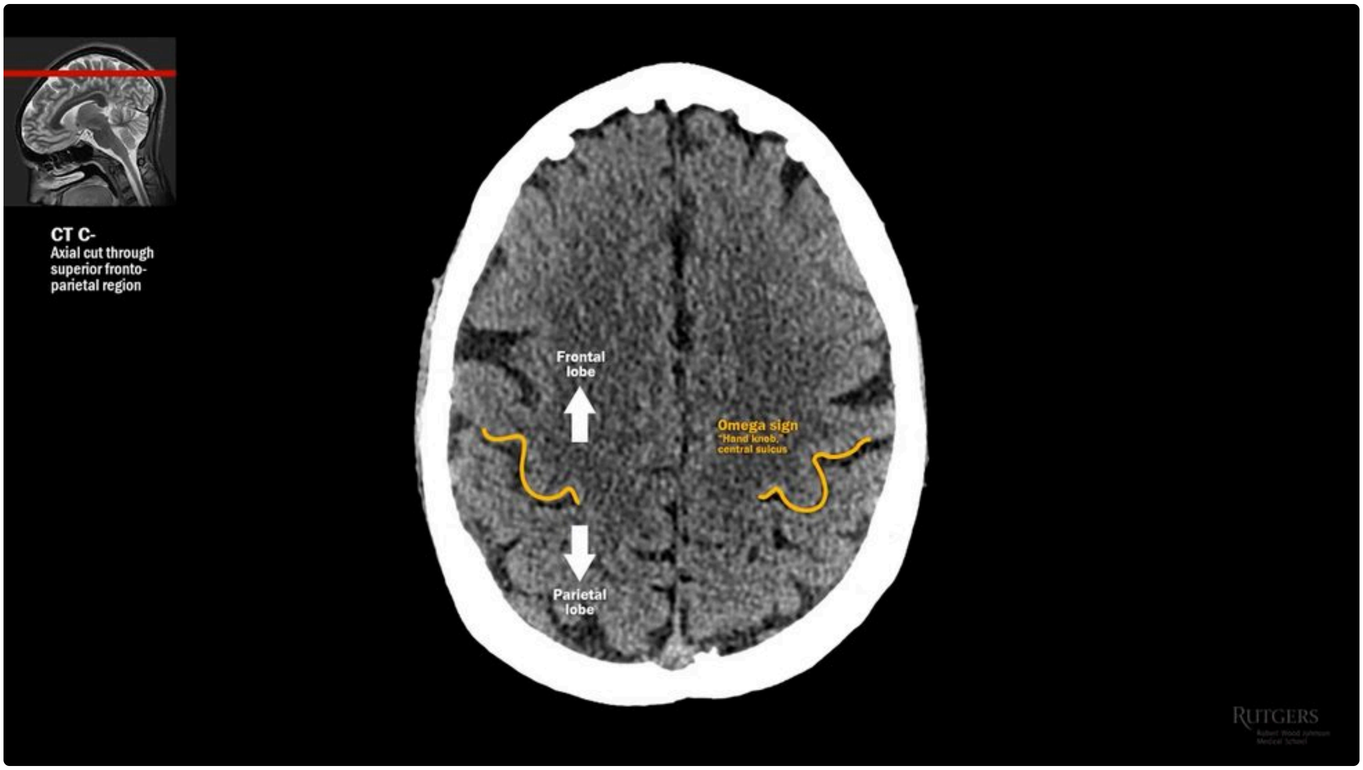
A normal axial head CT cut through the orbits. You can make out the lenses inside the eyeballs.

ORIENTATION, BY CONVENTION

The patient lies in front of you, feet first, as in the anatomy lab or the OR. **The nose points up. The patient's left is on your right.** The back of the head is at the bottom.

04 Landmarks By Level

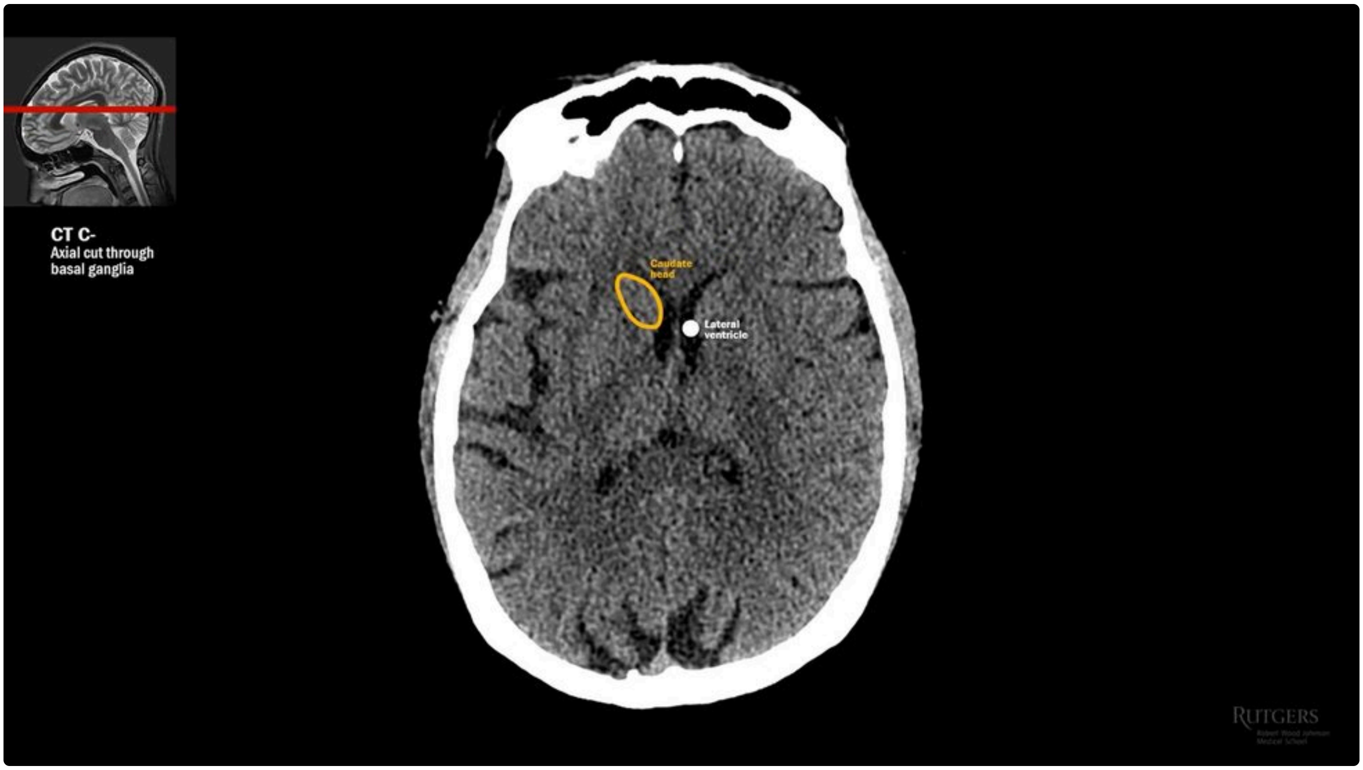
You cannot call something abnormal until you can name what is normal. Work down the brain, level by level. T1 is shown alongside because it is the anatomical sequence, showing white matter white and grey matter grey, as on gross pathology.



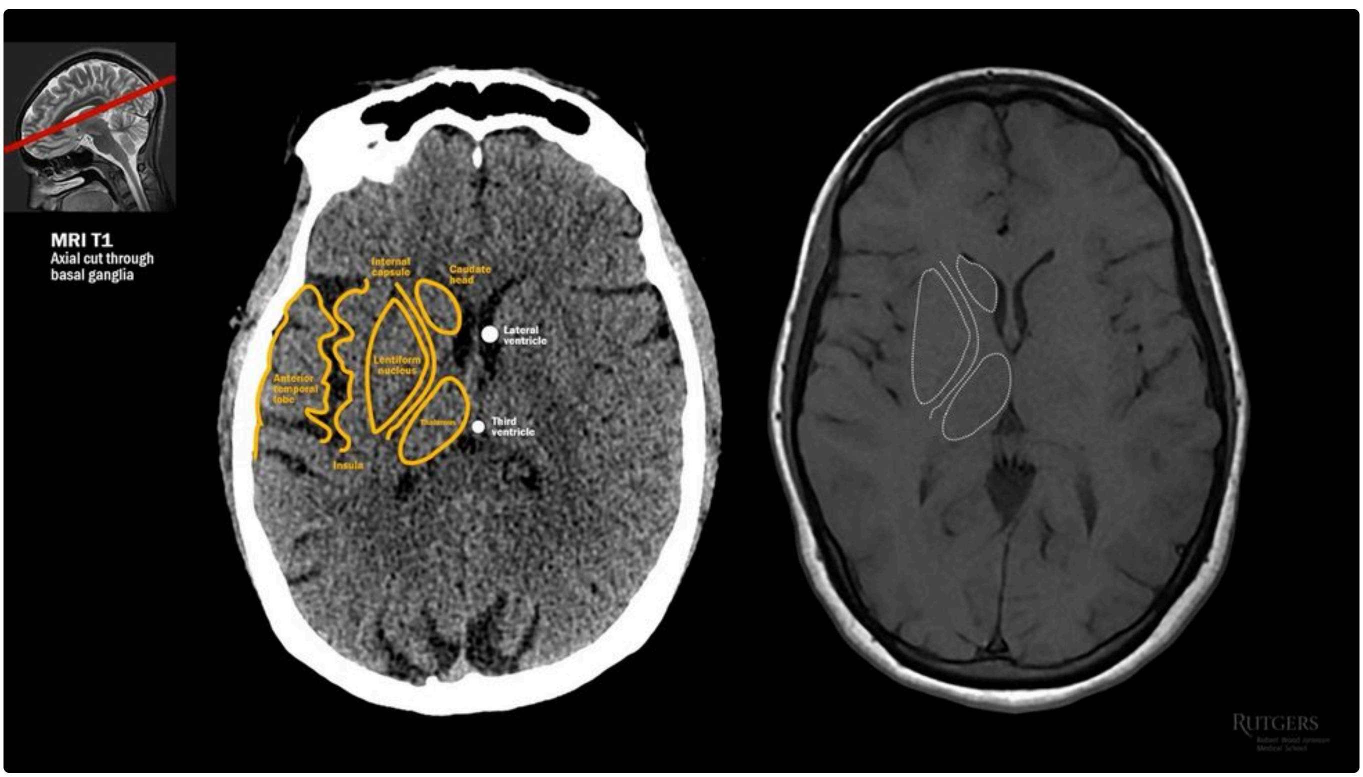
Upper frontal and parietal lobes. Find the central sulcus: it curves like the Greek letter omega.

THE HAND KNOB

That omega-shaped bump in the pre-central gyrus is the **hand knob**, where motor control of the hand lives. Anterior to the central sulcus is frontal lobe; posterior is parietal.



Basal ganglia level. Lateral ventricle centre, caudate head beside the frontal horn, thalami split by the third ventricle.



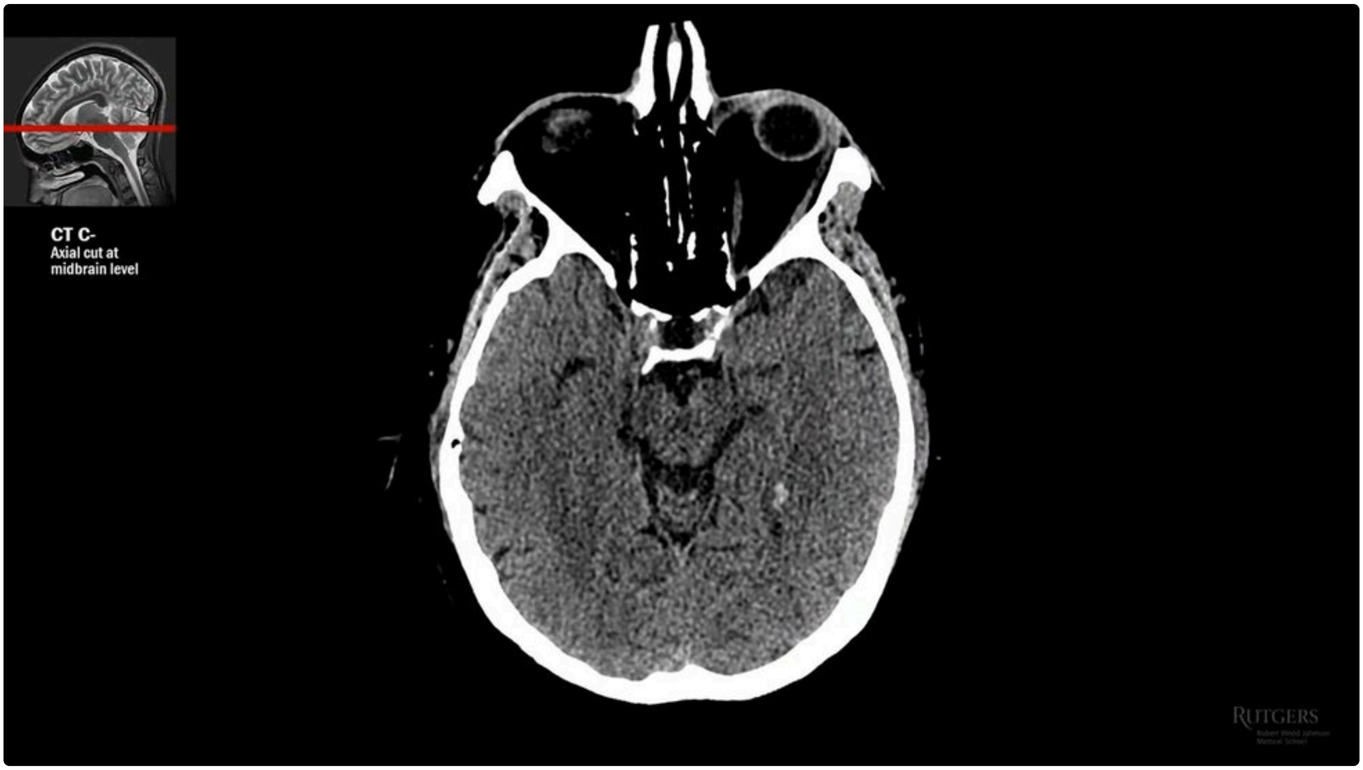
The same level on T1, with the illustration. Caudate, thalamus, internal capsule, lentiform nucleus, insula, anterior temporal lobe.

LEVEL	WHAT TO FIND
Superior frontoparietal	Central sulcus, omega sign, hand knob
Basal ganglia	Caudate head, thalamus, internal capsule, lentiform nucleus, insula
Sylvian fissure	Middle cerebral artery, orbits and globes return
Midbrain	Midbrain, basal cistern behind it, uncus laterally
Mid-pons	Pons, brachium pontis, fourth ventricle, cerebellopontine angle
Medulla	Medulla, fourth ventricle, cerebellum
Foramen magnum	Cervical cord bathed in CSF

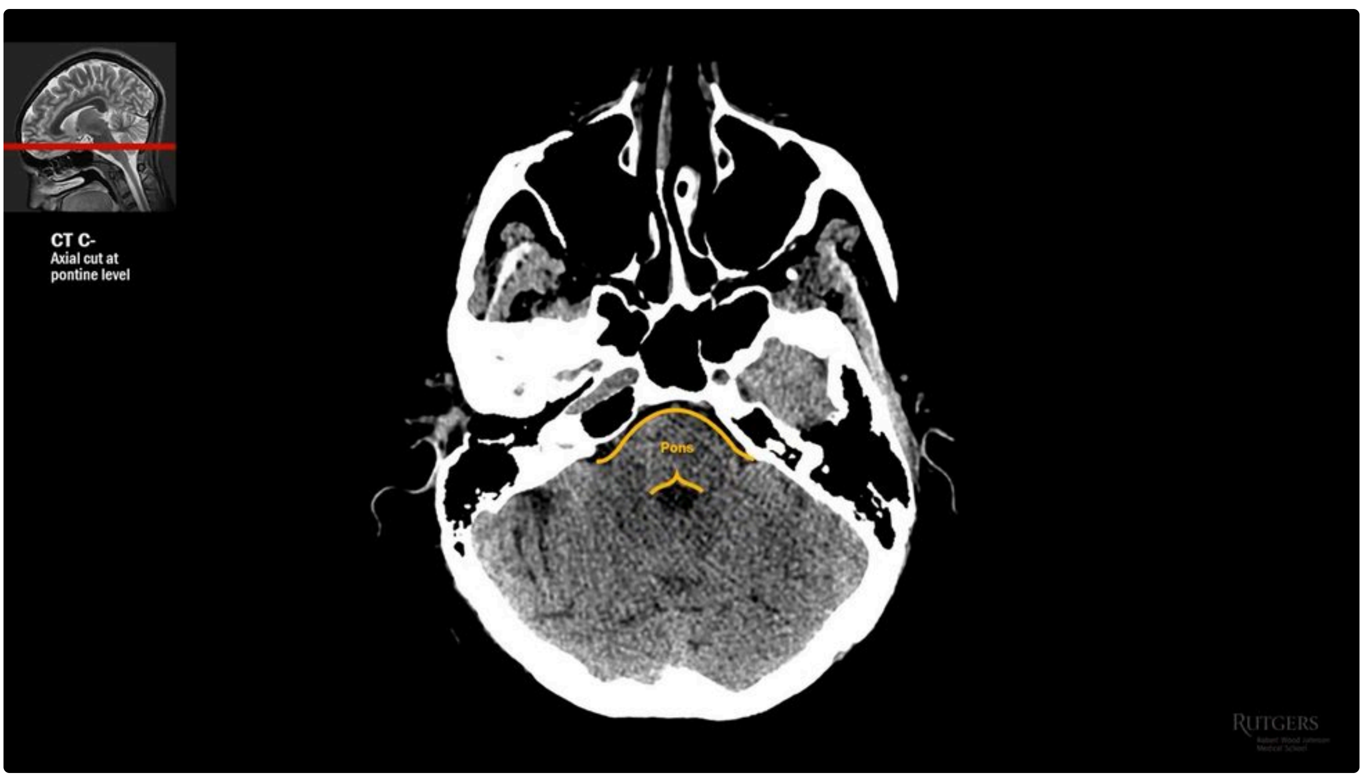
The Sylvian fissure. The MCA lives here, normally isodense to brain and hard to see.

WHEN YOU CAN SEE THE MCA

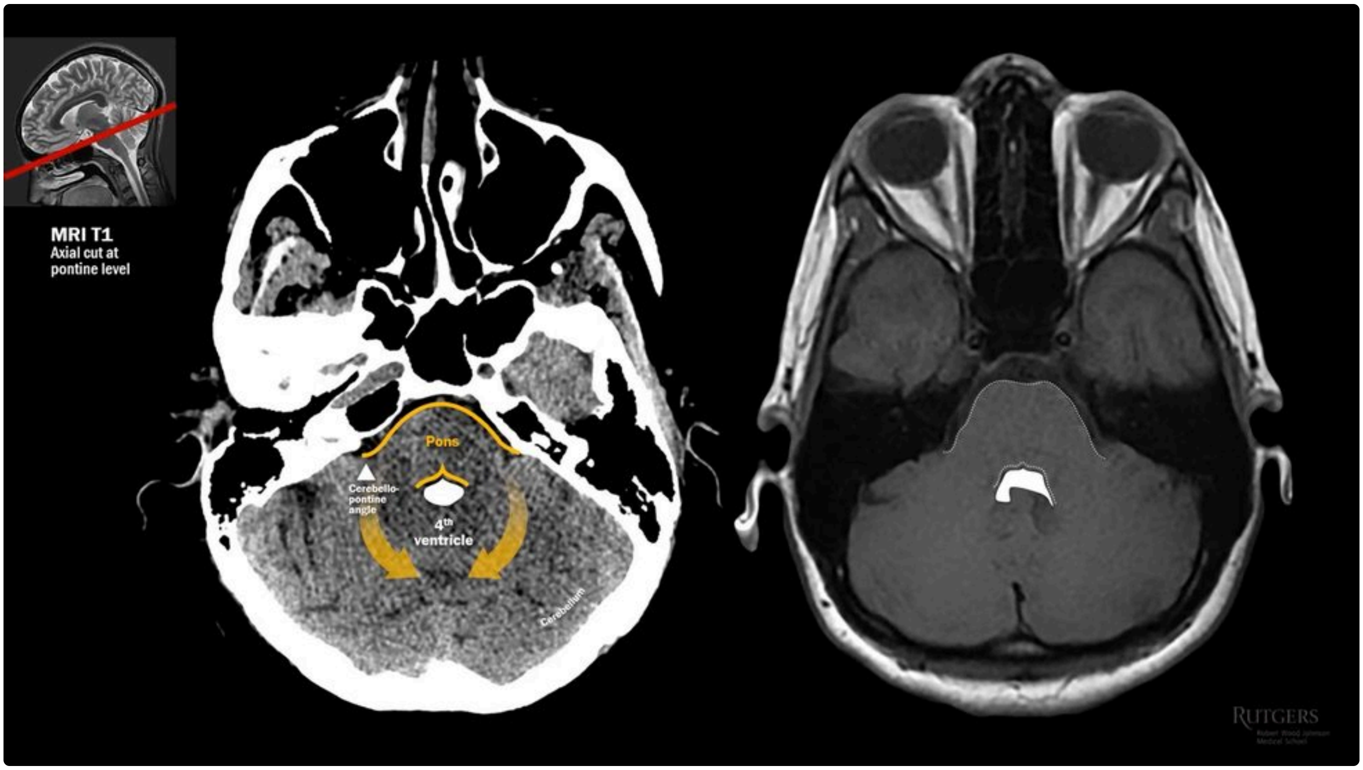
The middle cerebral artery is normally almost indistinguishable from nearby brain. **When you can make it out because it is brighter, things are bad** - that is a dense vessel sign, and it means acute clot.



Midbrain level. The uncus sits lateral to the midbrain, as in uncal herniation.



Mid-pontine level. The pons reaches out to the cerebellum with its brachia pontis, and the fourth ventricle sits in the middle of the hug.

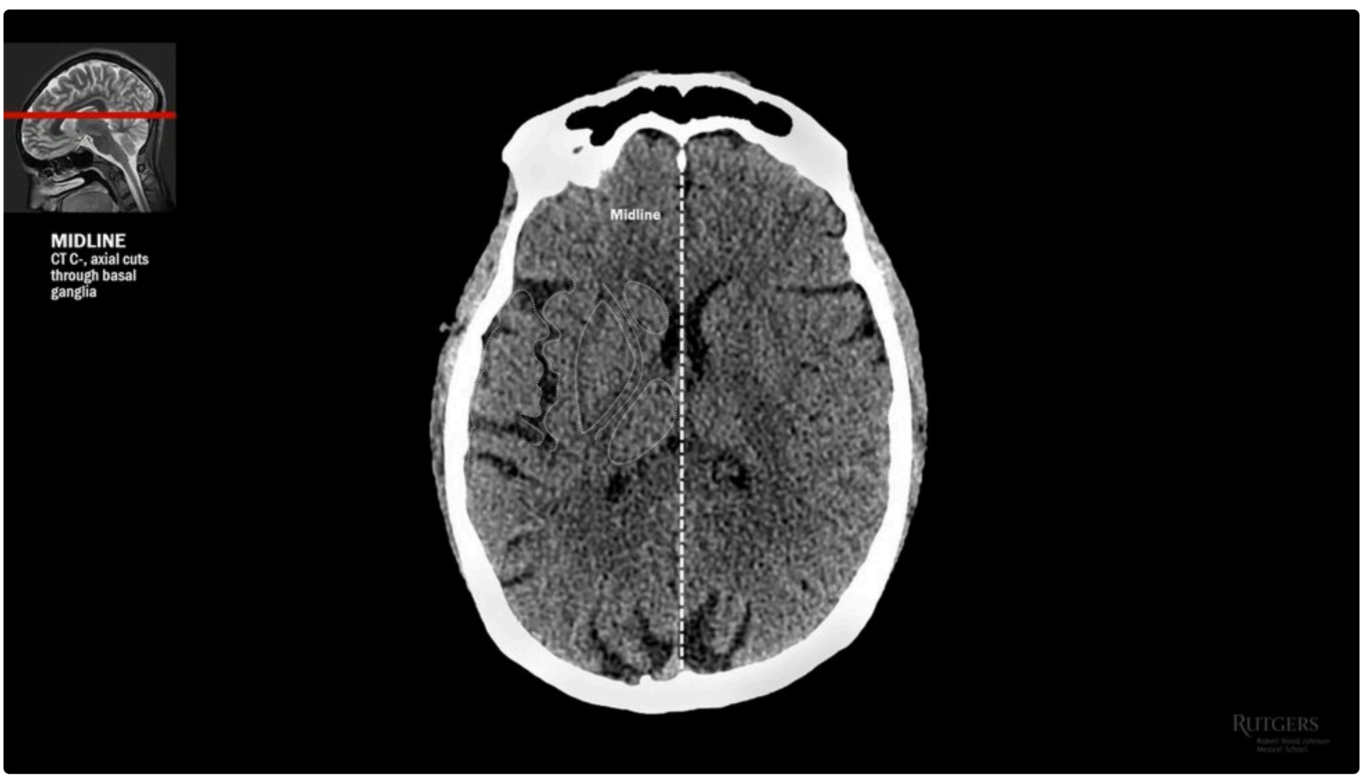


Medulla, fourth ventricle and cerebellum. Note how much better MR resolves the posterior fossa.

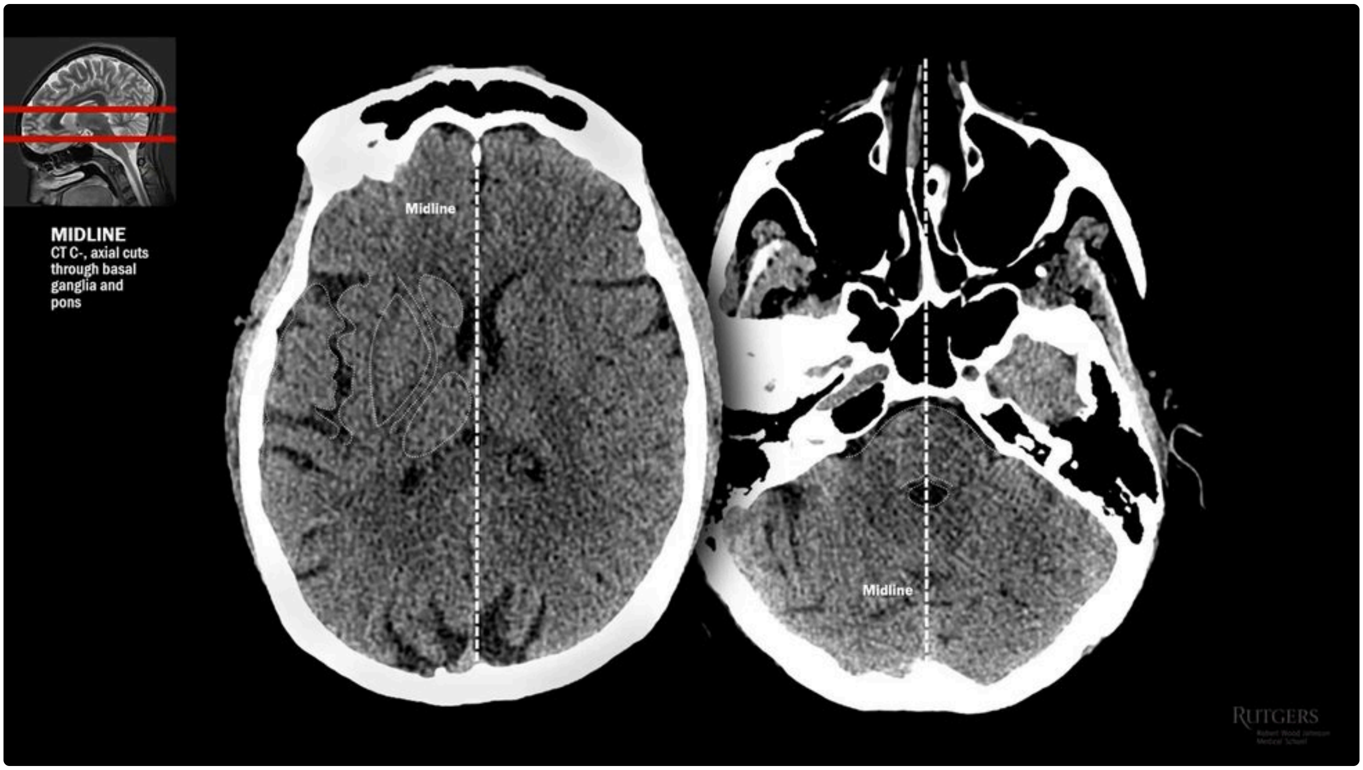
The temporal horn of the lateral ventricle is normally slit-like. **If it dilates, that is one of the earliest signs of hydrocephalus.**

05 Symmetry And Shift

Find midline first. Draw a line from the falx cerebri, through the septum pellucidum in the middle of the lateral ventricle, to the confluence of venous sinuses.



A symmetric study. Midline is where it should be.



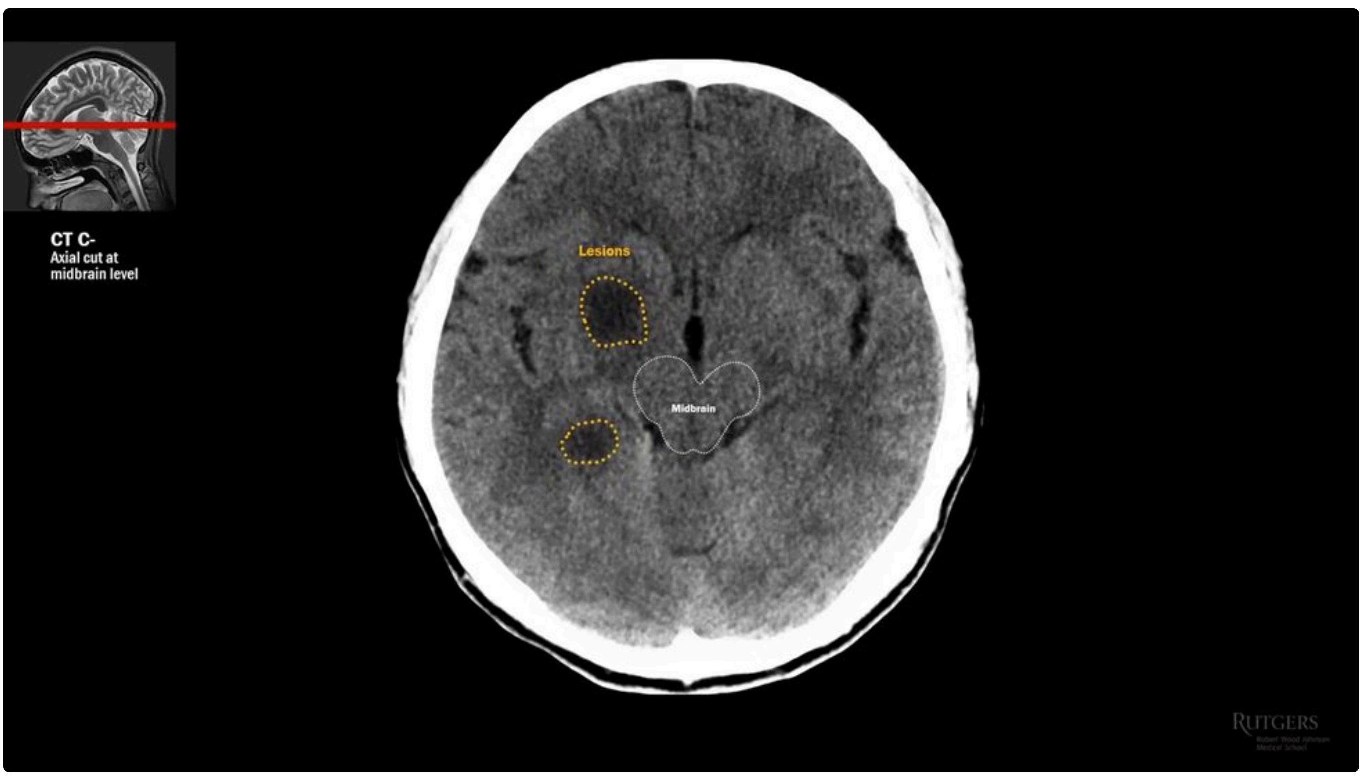
Asymmetric, with shift towards the left. The lesion must therefore be on the patient's right: a concave subacute subdural haematoma.

Also shifted, but the culprit here is a massive acute right MCA infarct.

THE ORDER MATTERS

Spot the asymmetry, **then** hunt the lesion causing it. **The shift points away from the lesion** - shift towards the left means the problem is on the right.

Posterior fossa, harder. Highlight pons, fourth ventricle and cerebellum and the shift appears: a vestibular schwannoma at the cerebellopontine angle.



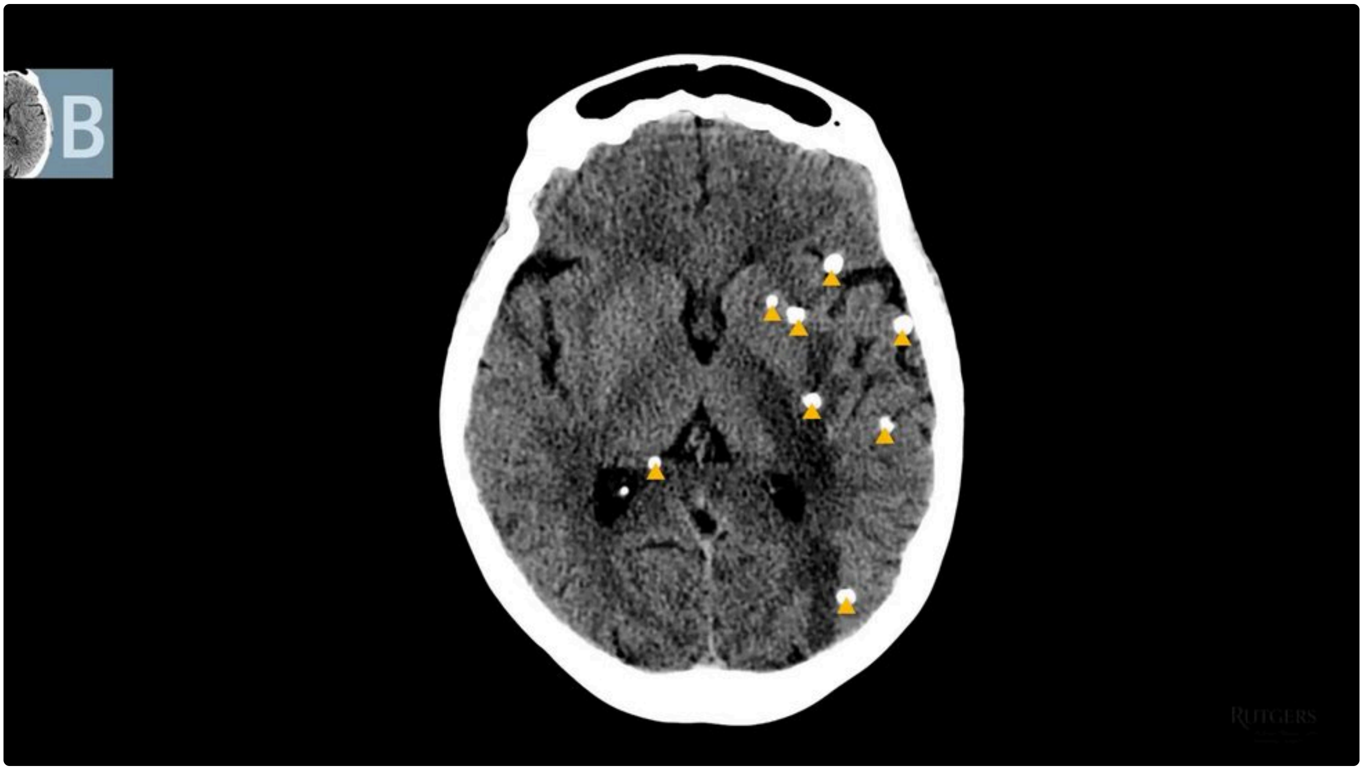
Not all asymmetry is midline shift. Two subcortical hypodensities on the right only - this patient has toxoplasmosis.

06 Bright On CT

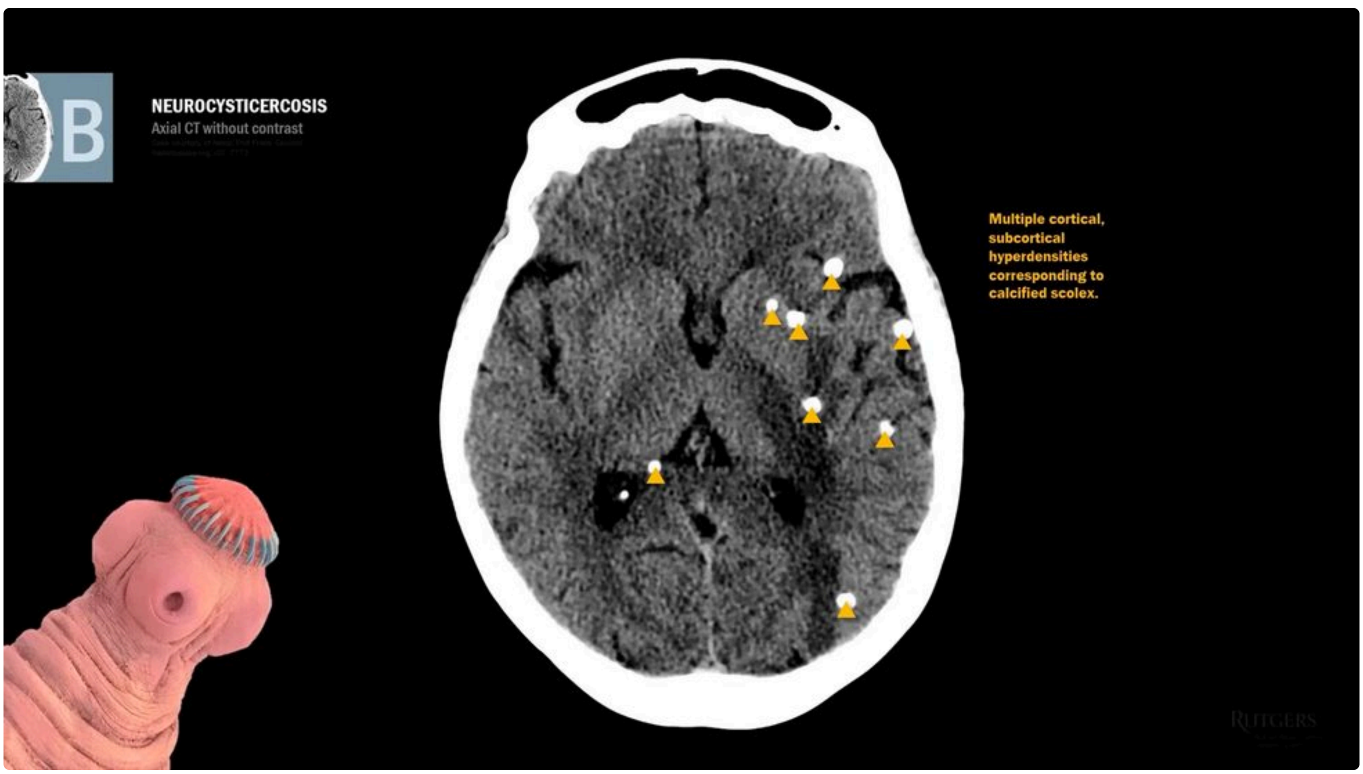
Hyperdensity is easy to spot, by design, which makes it an excellent diagnostic tool. **Head CT detects acute blood with over 90% sensitivity.**

CAUSE OF BRIGHTNESS	EXAMPLES
Mineralised structures	Bone, chronic calcified lesions
Normal calcification	Pineal, choroid plexus
Acute blood	Any compartment. This is the one that matters acutely.

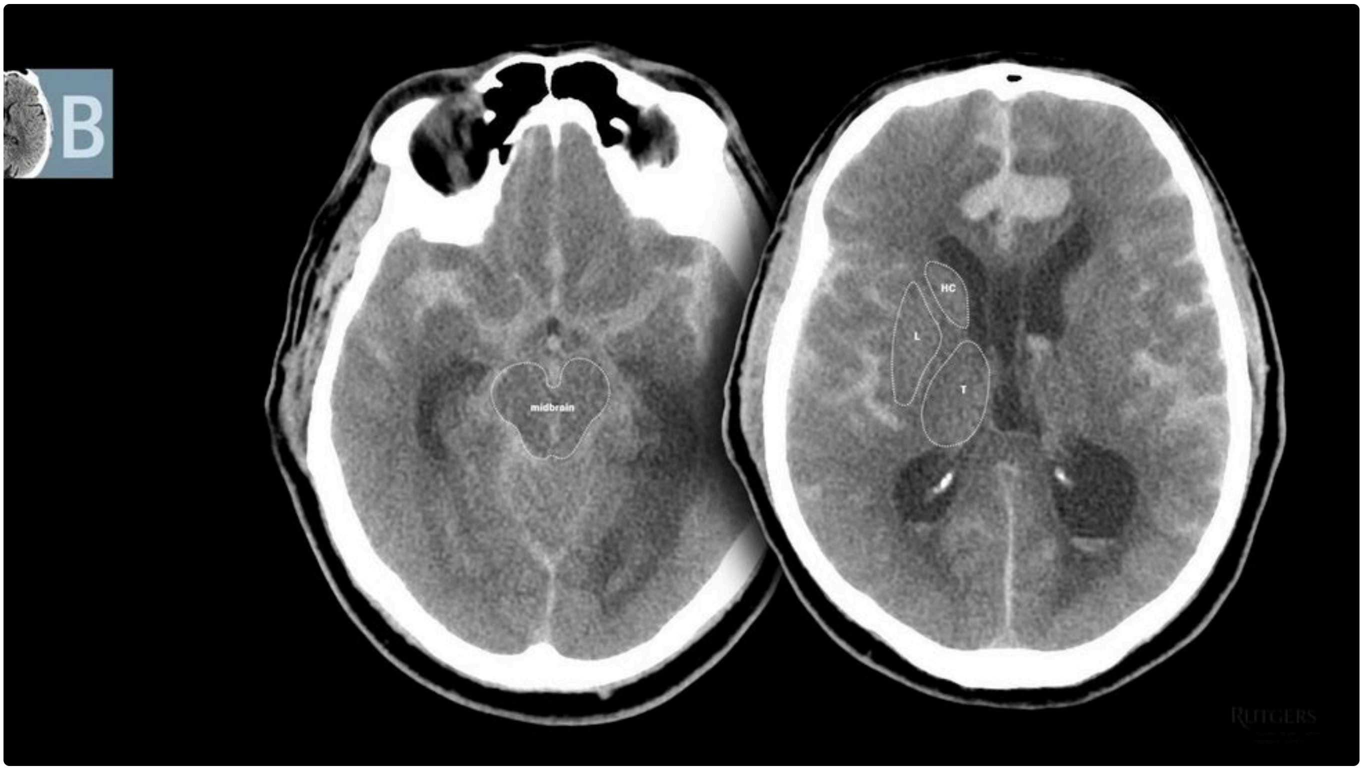
Not all brightness is abnormal: calcified pineal, calcified choroid plexus, bone.



Multiple calcified lesions - the calcified scolex of Taenia solium. Neurocysticercosis, a common cause of seizures worldwide.



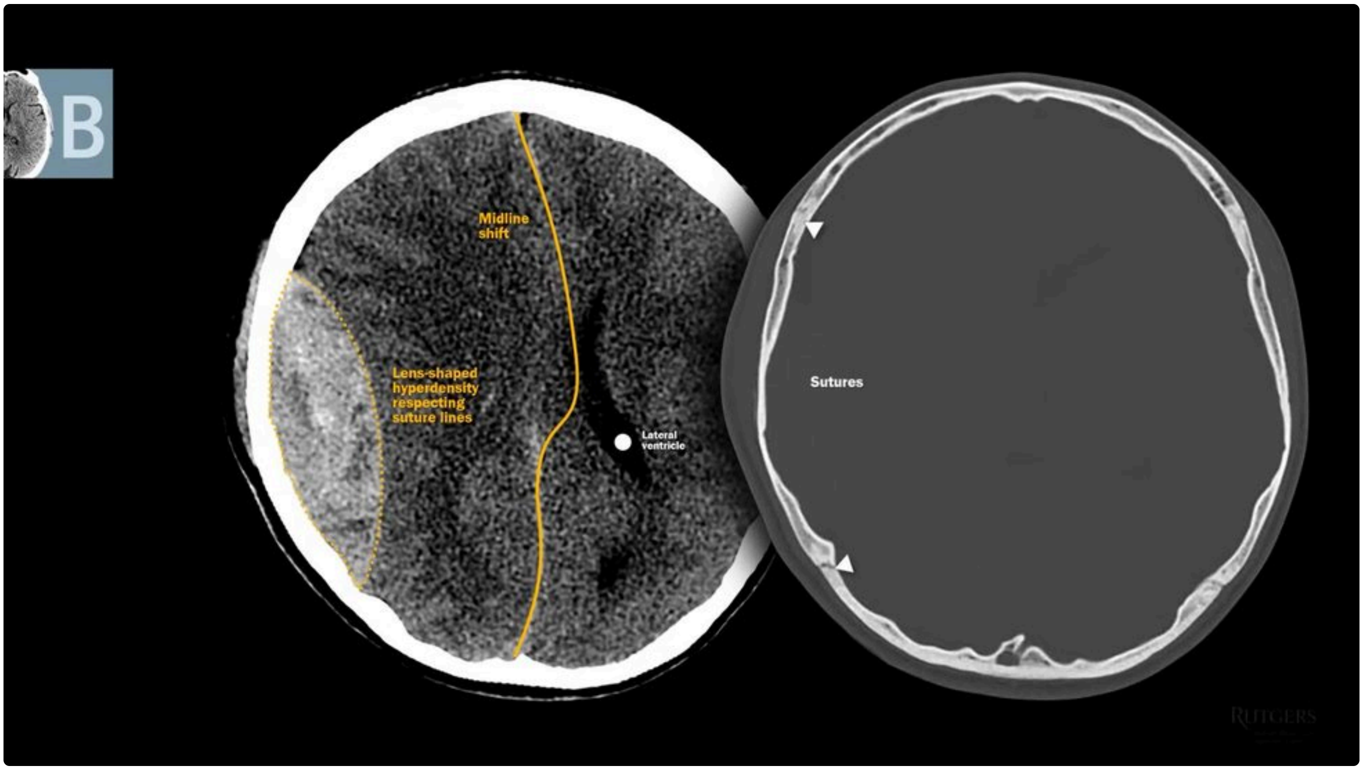
Basal ganglia hyperdensity with intraventricular extension: acute haemorrhage.



Sulci outlined as if with chalk. Diffuse subarachnoid haemorrhage, extra-axial.

SHAPE	COMPARTMENT	DIAGNOSIS
Lens-shaped (convex), respects suture lines	Extra-axial	Epidural haematoma
Concave, crosses suture lines	Extra-axial	Subdural haematoma

Lens-shaped, respecting sutures: epidural. Switch to bone window to see fractures and sutures.



Concave and crossing sutures: subdural.

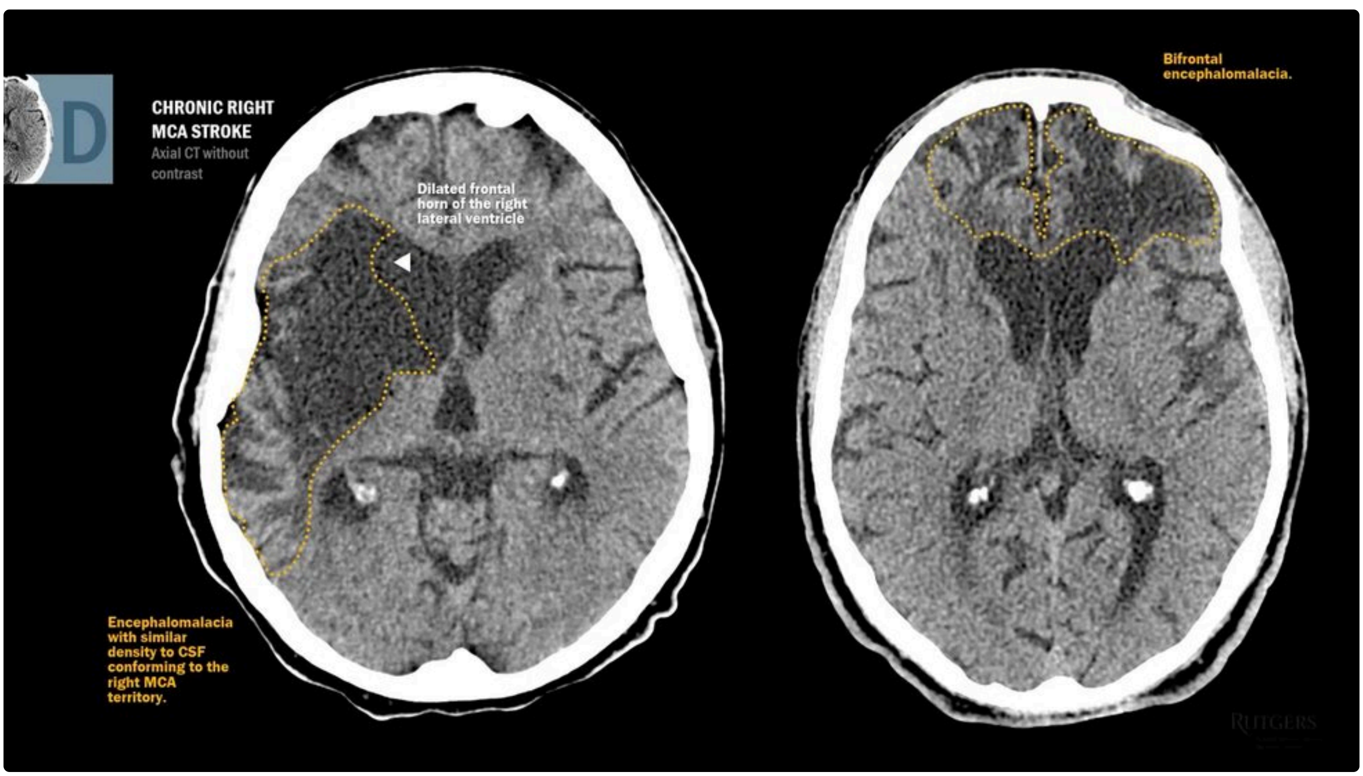
Two dense vessel signs. Left: serpentine hyperdensity in the Sylvian fissure, the MCA. Right: hyperdensity in the interpeduncular cistern, top of the basilar.

> **WHY DENSE VESSEL SIGNS MATTER**
>
> Acutely clotted blood is bright on CT. A dense vessel sign is therefore **a visible clot**, critical information in a suspected acute ischaemic stroke.

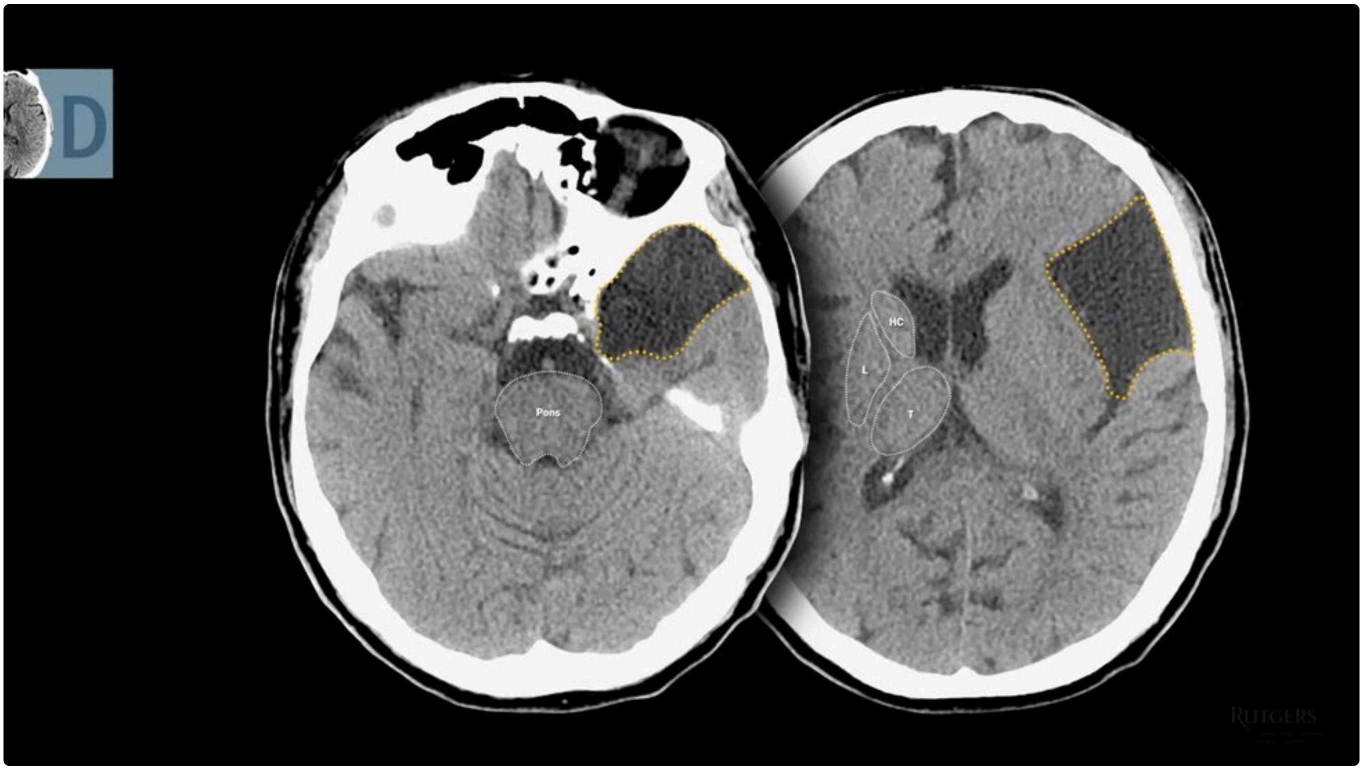
07 Dark On CT

The dark side is harder to read and holds more power. Fluid is less dense, so hypodensity means chronic lesions, cysts, or oedema.

Well-defined, wedge-shaped, CSF-density, conforming to a vascular territory. Chronic right MCA infarct with encephalomalacia and ex vacuo dilatation of the frontal horn.



Bifrontal encephalomalacia, not conforming to any vascular distribution. Bilateral frontal lobes are a classic trauma site: chronic traumatic brain injury.



Well-defined but smooth, extra-axial, CSF density. A frontotemporal arachnoid cyst.

> **THE 6 TO 8 HOUR PROBLEM**
>
> Ischaemic neurons swell immediately, but **that oedema takes roughly 6 to 8 hours to appear on CT**. Most early head CTs in acute stroke are normal. A normal CT does not exclude stroke.

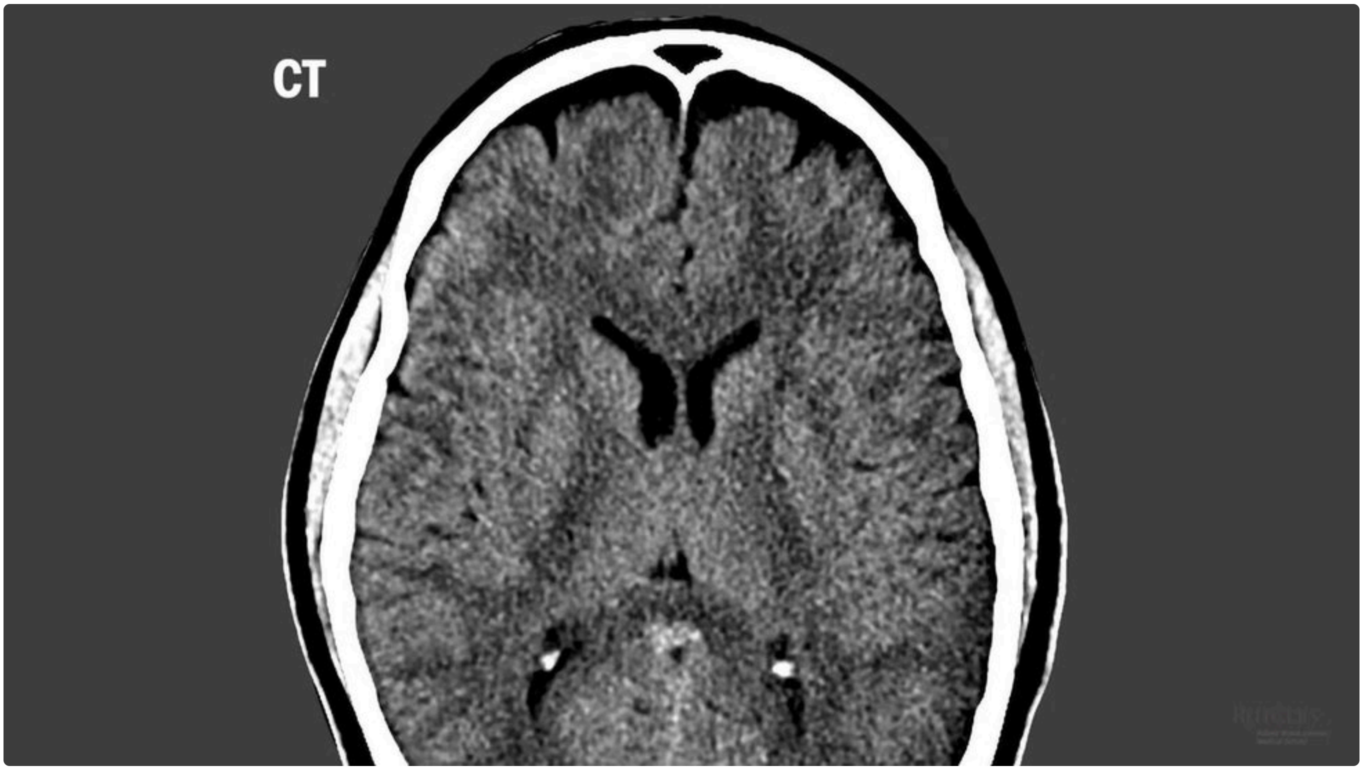
Faint hypodensity in the right MCA territory, 8 hours after symptom onset. Barely there.

Finger-like subcortical hypodensity, like a giant hand gripping a small ball. The ball is a metastasis; the hand is oedema of a different kind.

08 The MRI Bundle

Order a non-contrast MRI brain and you get a standard bundle. Know what each dish is for.

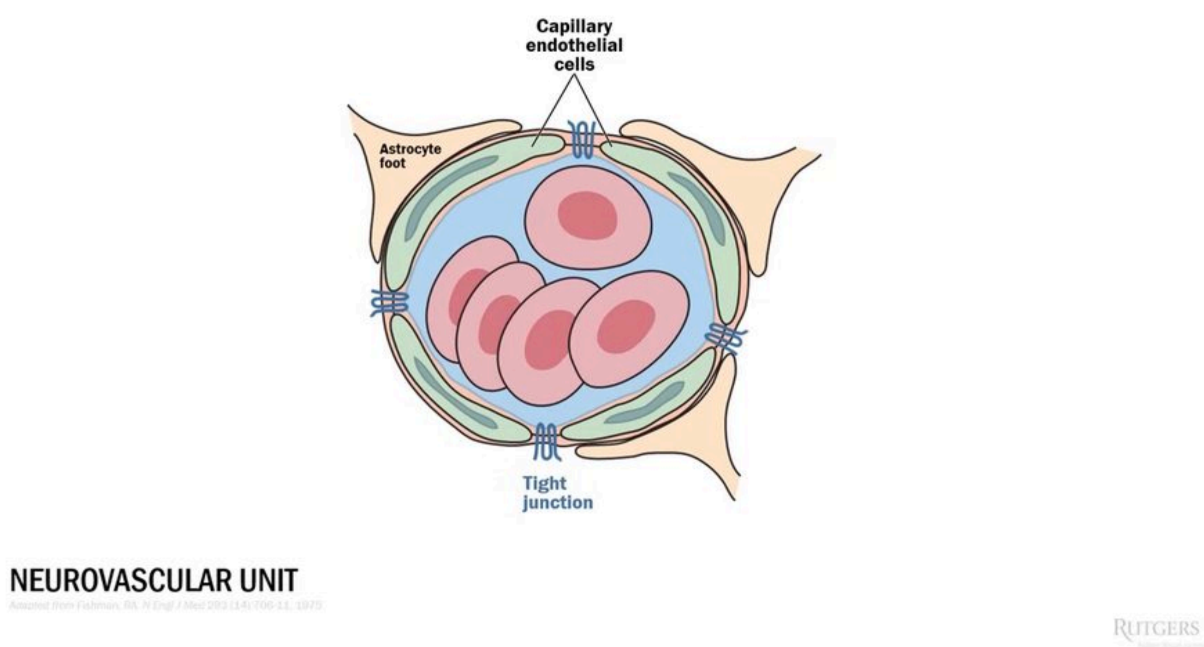
SEQUENCE	WHAT IT IS FOR
T1	Anatomy. White matter white, grey matter grey. Baseline for comparison.
T2	Fluid is bright. Excellent for oedema, but CSF is bright too.
FLAIR	T2 with the CSF signal subtracted. Oedema readable from across the room.
DWI	Energy failure and ischaemia, detectable from about 30 minutes.
ADC	DWI's counterpart. Confirms true restricted diffusion.
GRE	Haemorrhage.
T1 with gadolinium	Ordered separately. Blood brain barrier breakdown.
MRA and specials	Vasculature, fat suppression, perfusion, spectroscopy, tractography.



Zoom a normal CT and adjust contrast: denser grey matter, darker subcortical white matter, and the grey-white junction where they meet.

09 The Barrier

Two lines of basic science carry the entire second half of this guide. They are worth the minute.



The neurovascular unit: astrocyte foot processes wrapped around capillary endothelial cells joined by tight junctions.

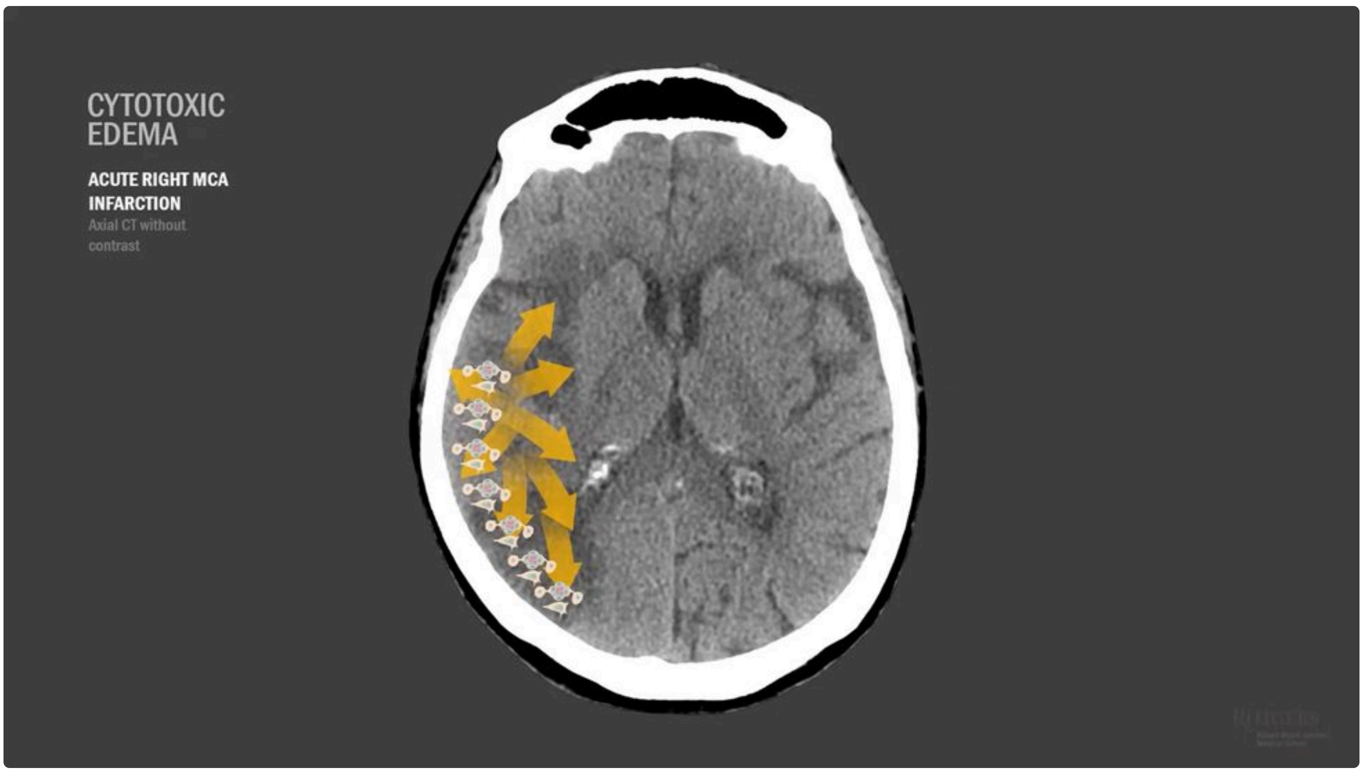
- **The neurovascular unit is the blood brain barrier**, and the barrier surrounds the entire brain parenchyma.

- **Gadolinium does not cross an intact barrier.** So you should never see enhancement in brain parenchyma or meninges when the units are healthy.

> **TWO WAYS TO BREAK IT, TWO KINDS OF OEDEMA**
> Starve the unit of blood and cells swell with the barrier still intact: **cytotoxic**. Damage the tight junctions so plasma leaks out: **vasogenic**. Everything downstream follows from which one you are looking at.

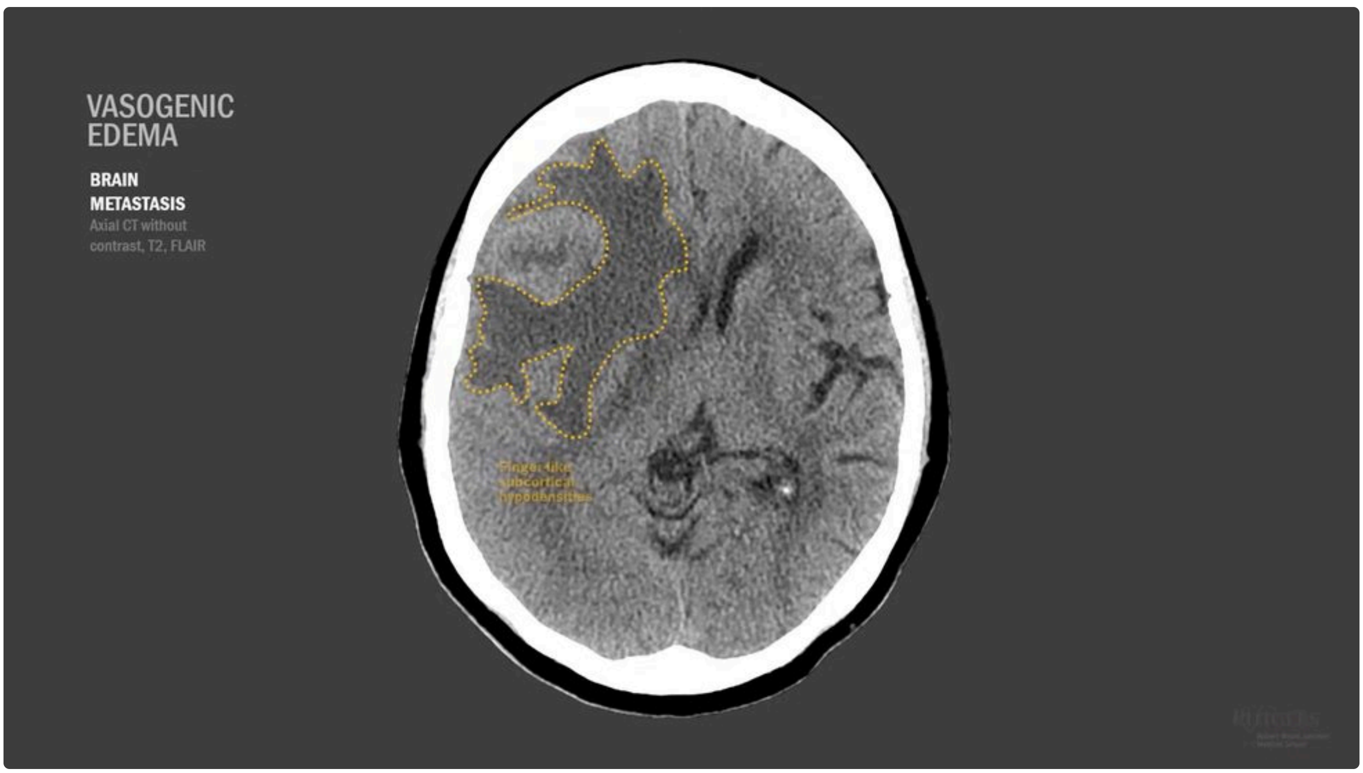
10 Two Kinds Of Edema

	CYTOTOXIC	VASOGENIC
Mechanism	Energy failure. Cells cannot hold the sodium-potassium gradient and take on fluid.	Tight junctions leak. Plasma escapes into interstitium.
Barrier	Intact early	Compromised
Causes	Ischaemia, stroke	Neoplasm, abscess, trauma, severe hypertension
Grey matter	**Involved.** Cortical ribbon lost.	**Spared.** Cortical ribbon preserved.
Shape	Follows a vascular territory	Finger-like subcortical projections
Grey-white junction	Lost	Oedema stops at it



Acute right MCA infarct. Oedema in both grey and white matter; the cortical ribbon is gone. That loss is your clue.

The same patient on CT, T2 and FLAIR. On T2 the oedema is bright but so is CSF. FLAIR subtracts the CSF and the abnormality jumps out.



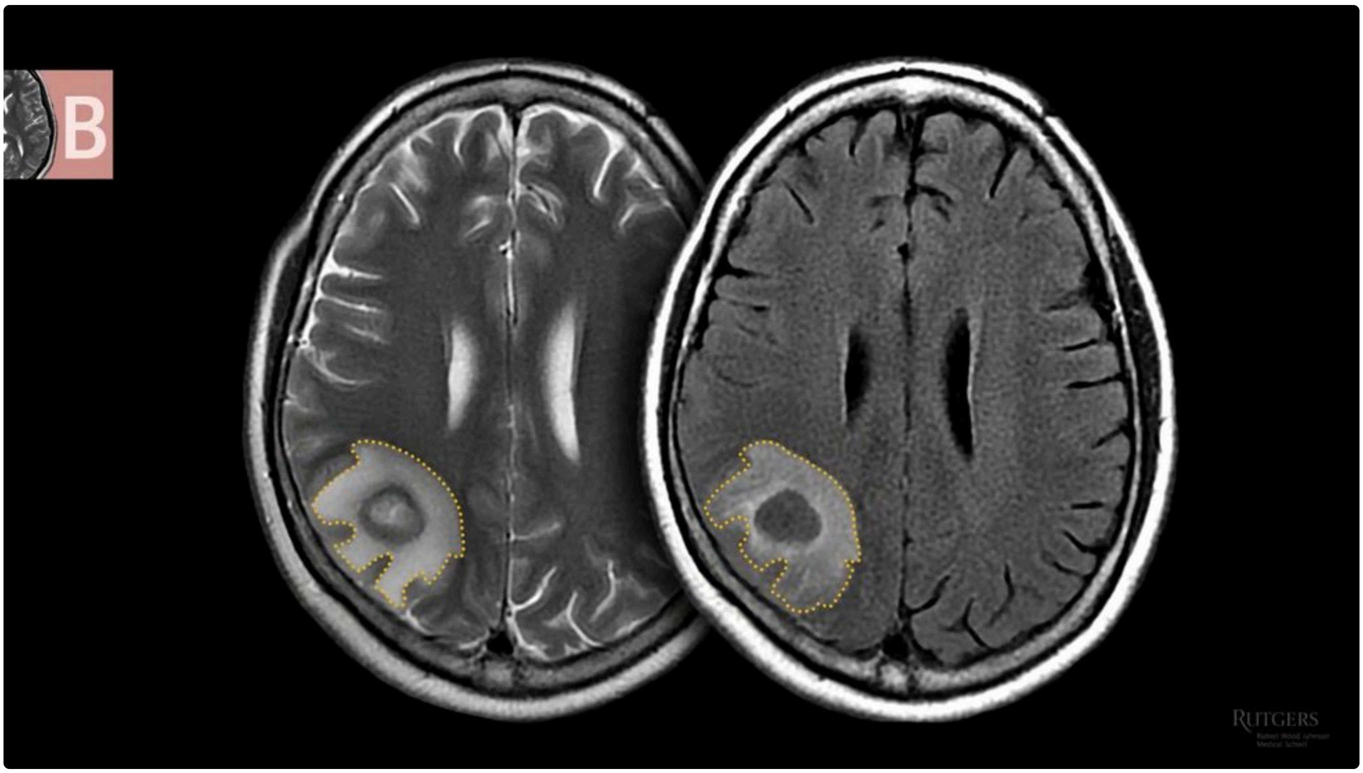
A brain metastasis. Finger-like projections stopping at the grey-white junction, cortical ribbon preserved, heterogeneous mass in the middle.

> **THE SINGLE MOST USEFUL DISCRIMINATOR**
>
> **Look at the cortical ribbon.** Lost means cytotoxic, so think ischaemia. Preserved, with finger-like white matter projections, means vasogenic, so think mass, abscess, trauma or hypertension.

11 Bright On T2

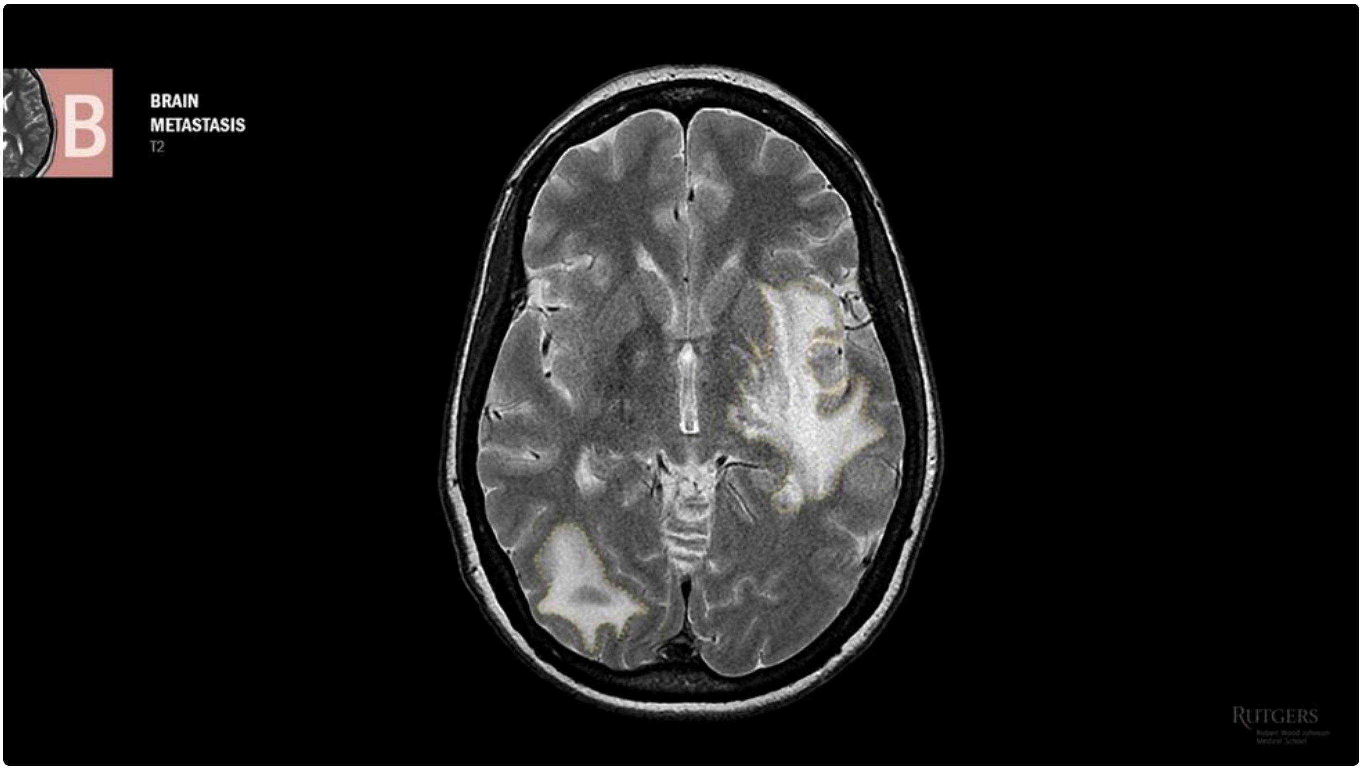
MRI is designed to highlight lesions, so most abnormalities are hyperintense. If MRI were a book, chapter one would be titled "look on the bright side, it is swollen."

BRIGHT ON T2 / FLAIR	EXAMPLE
Cytotoxic oedema of ischaemia	Acute infarct
Vasogenic oedema	Metastasis, abscess
Transependymal CSF flow	Obstructive hydrocephalus
Inflammatory lesions	Multiple sclerosis
Small vessel ischaemic change	Vascular dementia
Subacute haematoma	3 to 14 days



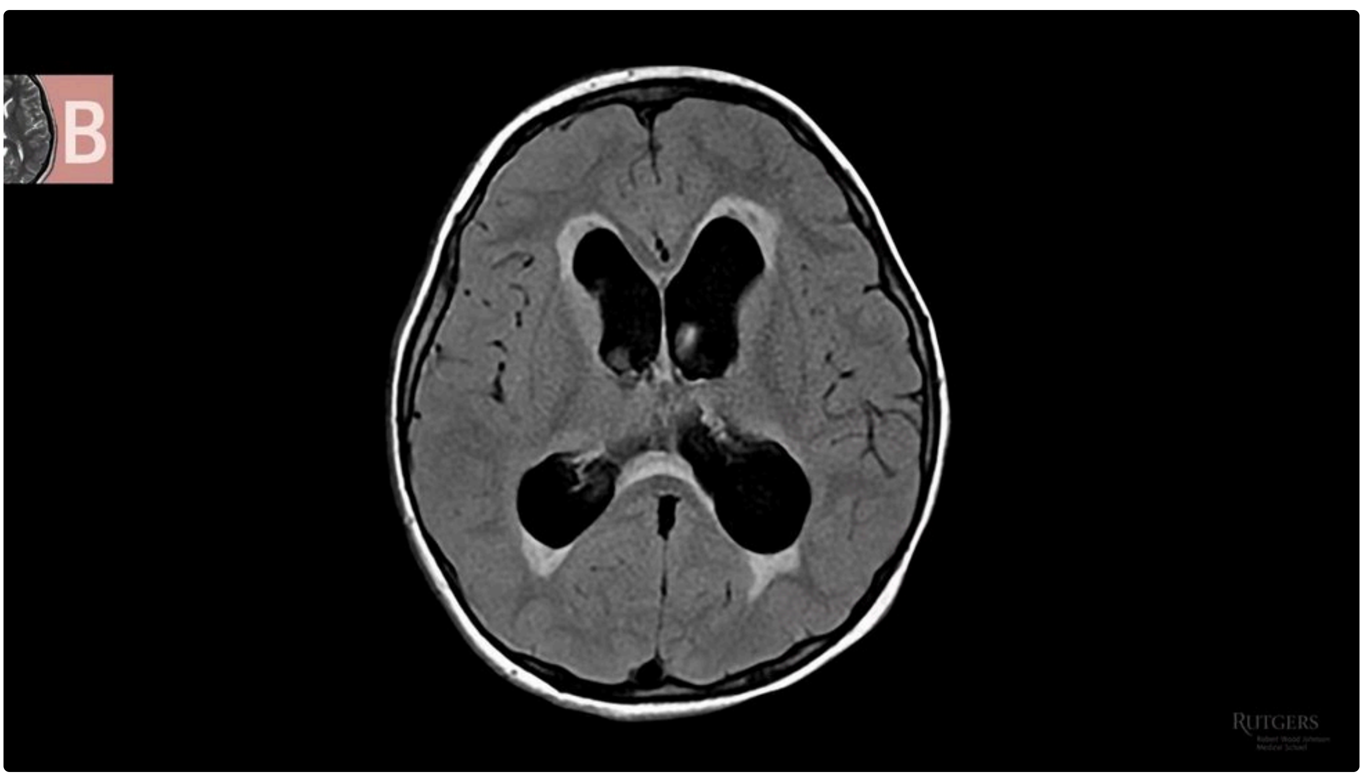
Finger-like projections, grey matter unaffected: vasogenic. A brain abscess in the right parietal lobe.

Abscess-forming organisms usually arrive through the bloodstream, so most emboli land in the territory of the artery supplying most of the hemisphere. **Frontal and parietal lobes, MCA distribution**, is therefore the commonest site.

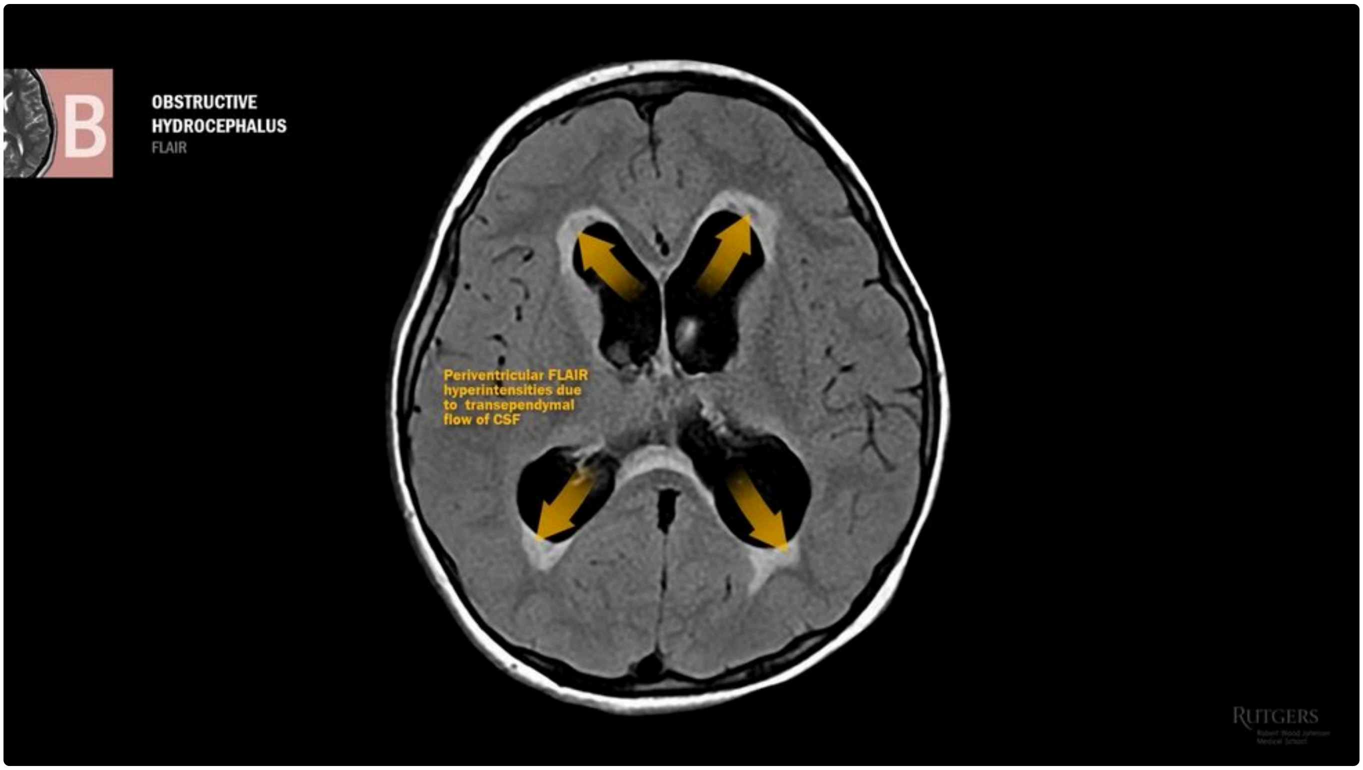


Finger-like vasogenic oedema around small, well-defined, round masses isointense to brain. Metastasis is the most common brain neoplasm.

> **WHY METASTASES SIT WHERE THEY DO**
>
> **Nearly 80% of metastases are at the grey-white junction.** Arterioles there are just small enough to trap travelling neoplastic cells. Lung and breast primaries are commonest, but imaging is never specific enough - tissue confirms the type.

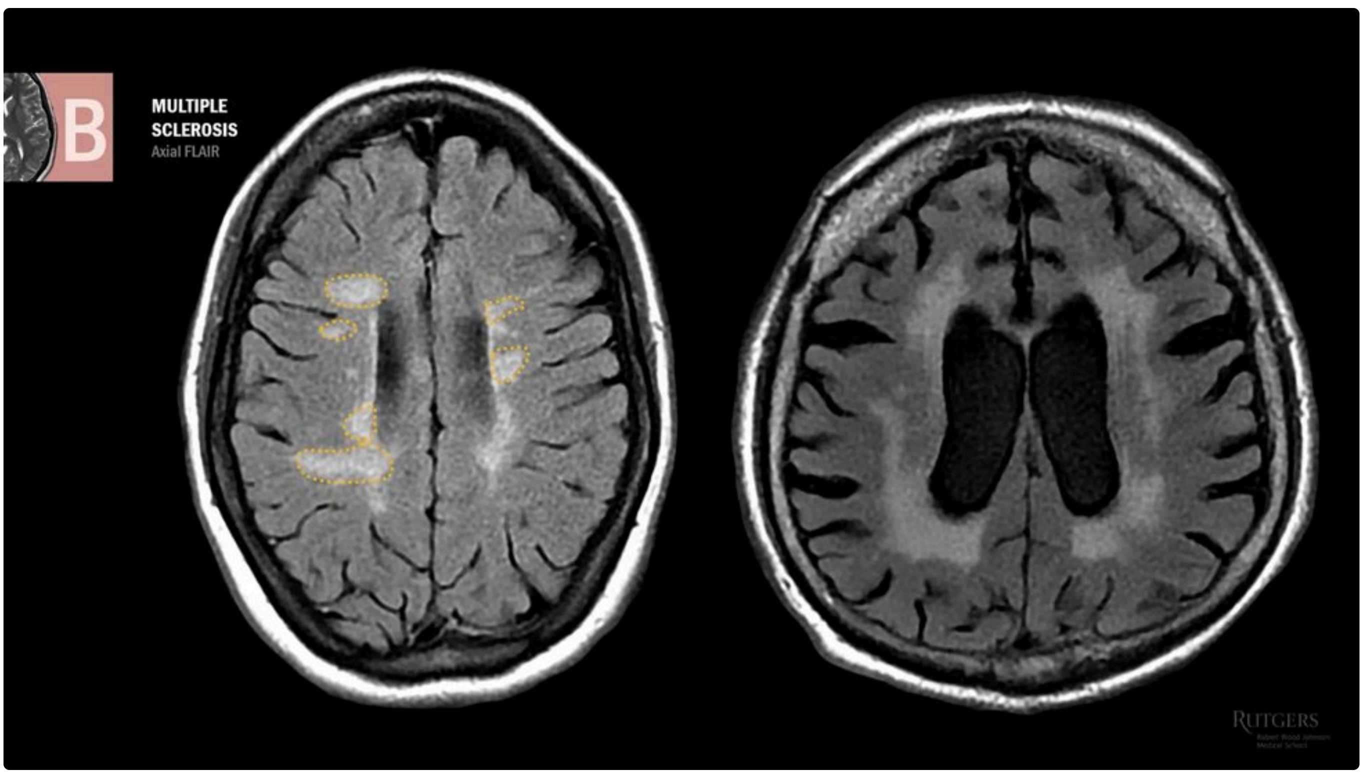


Symmetric smooth periventricular hyperintensity with dilated ventricles. Under pressure CSF is forced across the ependyma: transependymal flow in obstructive hydrocephalus.



Ovoid periventricular hyperintensities: Dawson's fingers in multiple sclerosis.

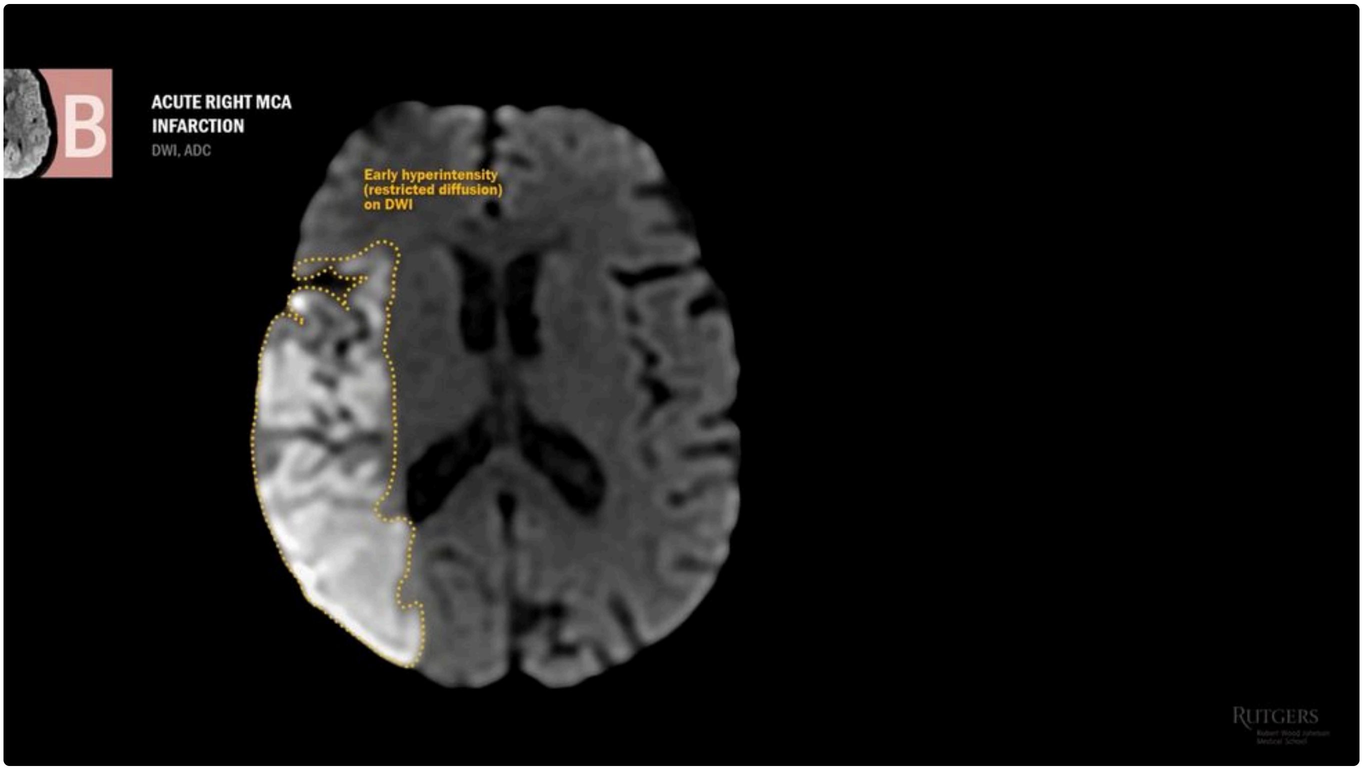
Why ovoid? Central veins run perpendicular to the ventricles. In MS, immune cells exit these veins and penetrate the parenchyma along them, producing that strangely specific shape.



Confluent rather than ovoid, with deep prominent sulci from atrophy. Small vessel ischaemic disease in a patient with vascular risk factors.

12 DWI And Its Twin

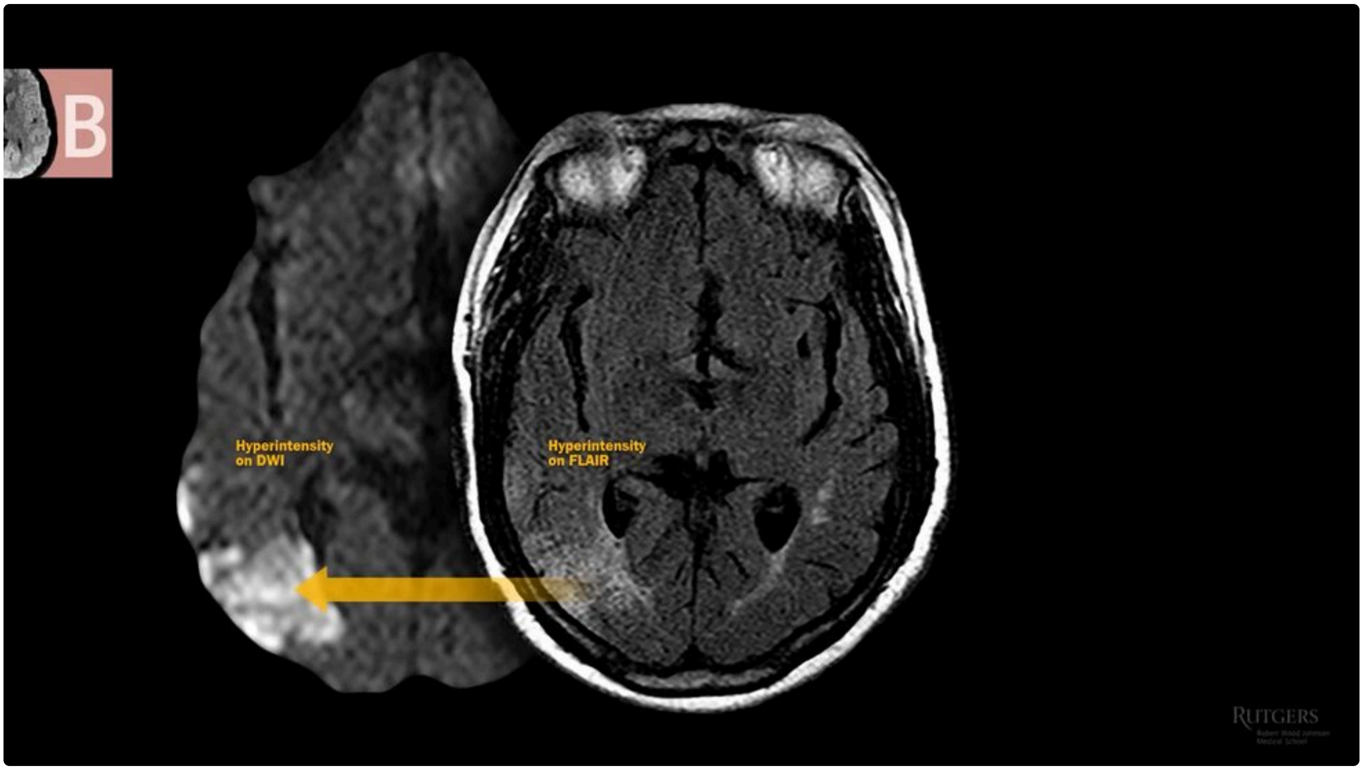
T2 and FLAIR behave like CT for ischaemia: it takes hours to build enough oedema to show. **DWI detects it within about 30 minutes.**



Early hyperintensity with restricted diffusion on DWI, well before T2 and FLAIR would show it.

ALWAYS CHECK DWI AGAINST ADC

True acute ischaemia is bright on DWI and DARK on ADC. If DWI is bright and ADC is also bright, you are looking at T2 shine-through, not a stroke.



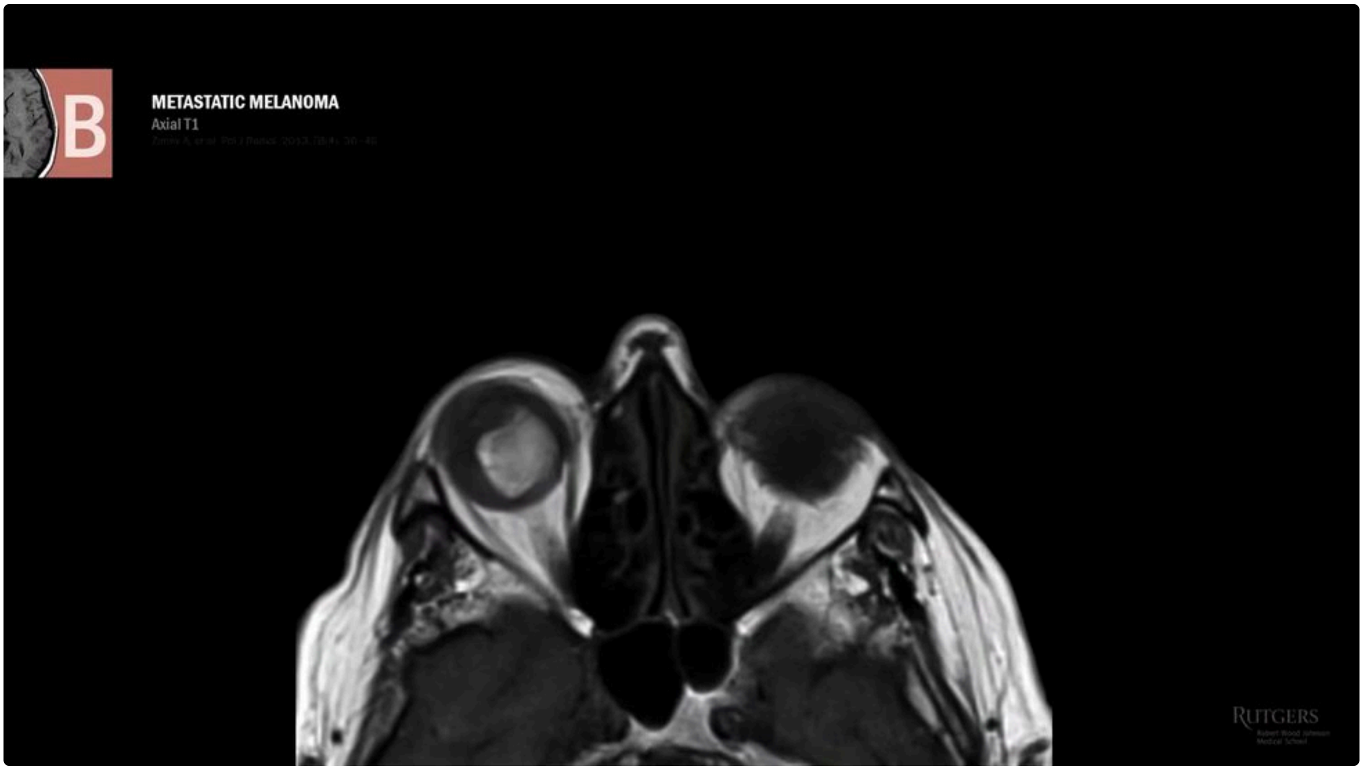
Bright on DWI, but bright on ADC too. This is T2 shine-through: whatever is bright on T2 and FLAIR shines through to DWI.

DWI is also highly sensitive and specific for **brain abscess**, which is how you separate an abscess from a necrotic tumour when ring enhancement alone will not settle it.

13 Bright On T1

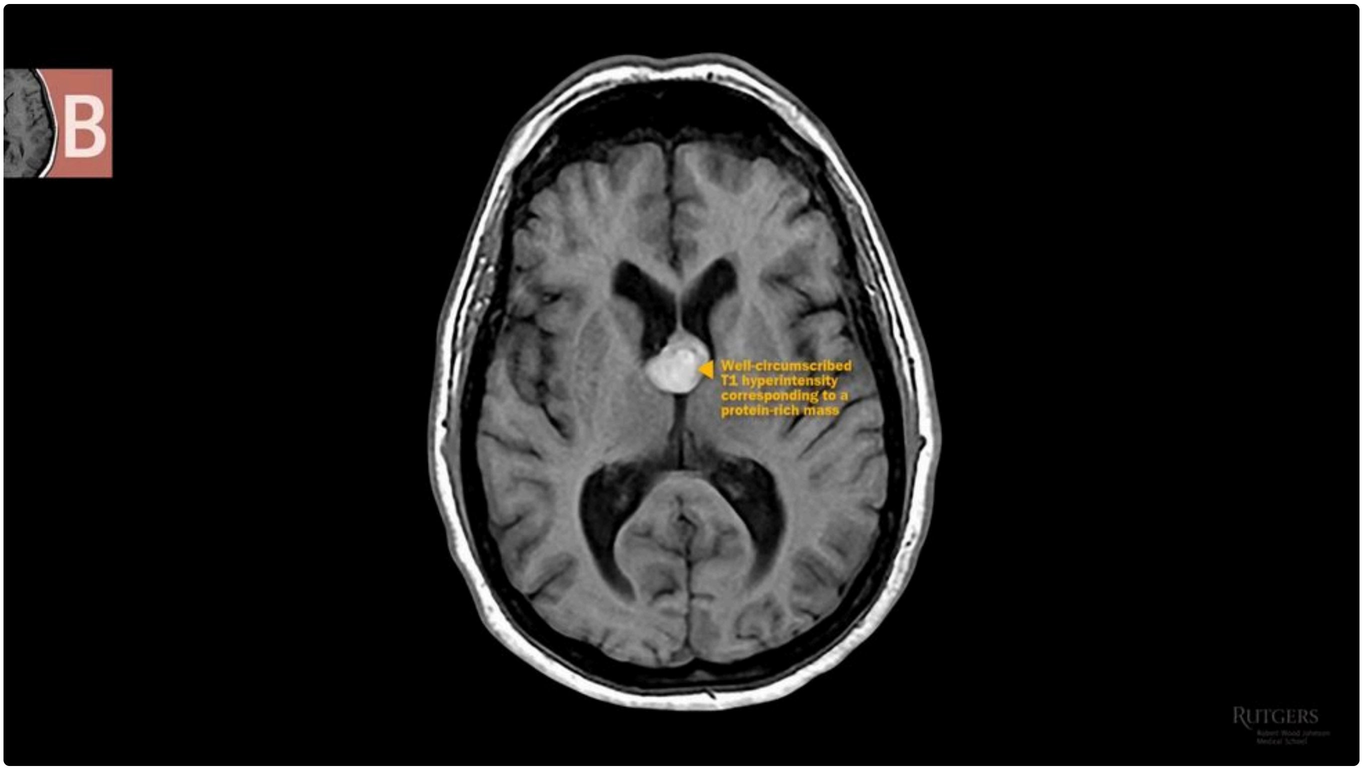
T1 brightness is a much shorter list, and a much more specific one.

BRIGHT ON T1	EXAMPLE
Paramagnetic substances	Iron, copper, melanin, calcium
Fat	Normal orbital fat
Protein-rich lesions	Colloid cyst
Subacute bleed	3 to 14 days



Metastatic melanoma in the right globe, T1 hyperintense. Note orbital fat is also bright, which is normal.

T1 hyperintensity in the basal ganglia from pathological copper accumulation: Wilson's disease.

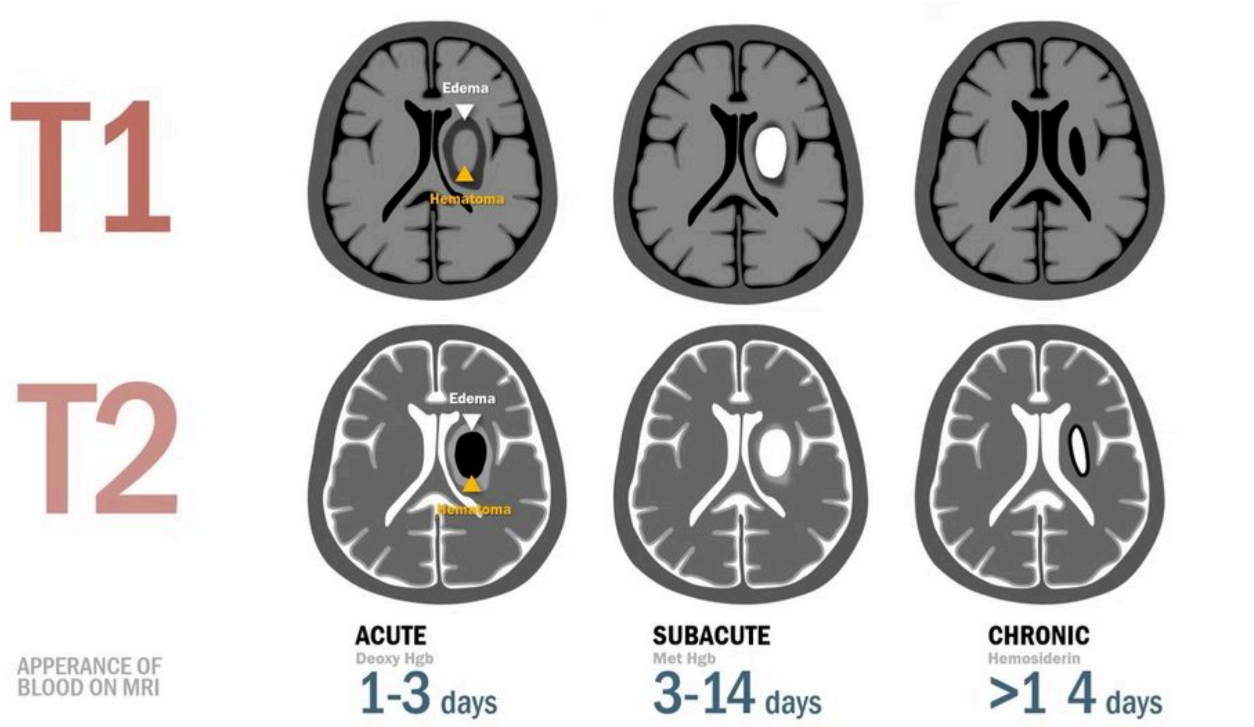


A well-circumscribed protein-rich T1 hyperintensity at the foramen of Monro: colloid cyst.

> **WHY A COLLOID CYST IS INFAMOUS**
> It can act as a **ball valve**, occluding the foramen of Monro and causing hydrocephalus so acutely and so severely that it kills.

14 Blood Over Time

Blood does not play by the rules. Its signal **evolves**, so the same haemorrhage looks different depending on when you scan it.



The appearance of blood on MRI across acute, subacute and chronic stages.

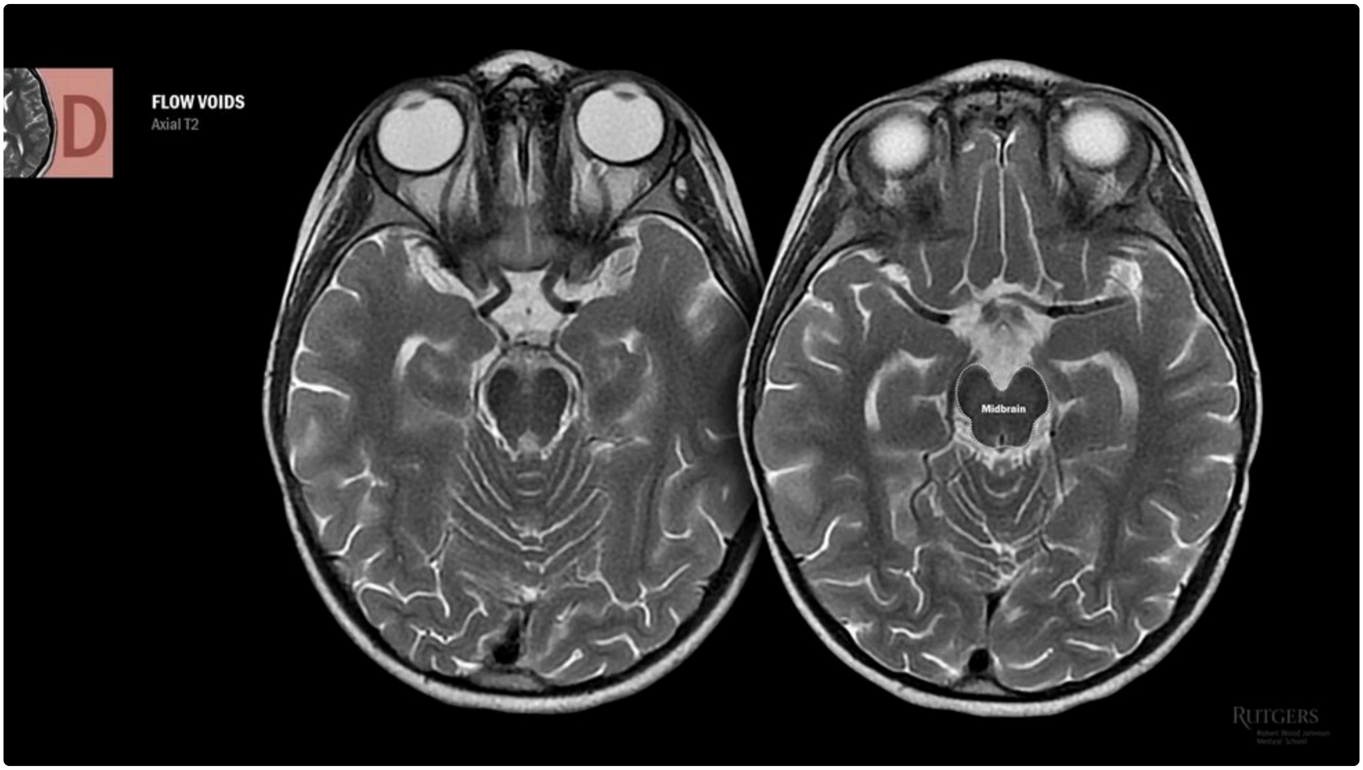
STAGE	TIMING	T1	T2
Acute	1 to 3 days	Isointense	**Dark** (deoxyhaemoglobin)
Subacute	3 to 14 days	**Bright**	**Bright** (methaemoglobin)
Chronic	beyond about 14 days	Dark	Bright, with a dark hemosiderin rim

Ultimately the site is replaced by a CSF-filled slit, dark on T1 and bright on T2, as if the brain had been stabbed. **The hemosiderin rim is the fingerprint left behind by old haemorrhage.**

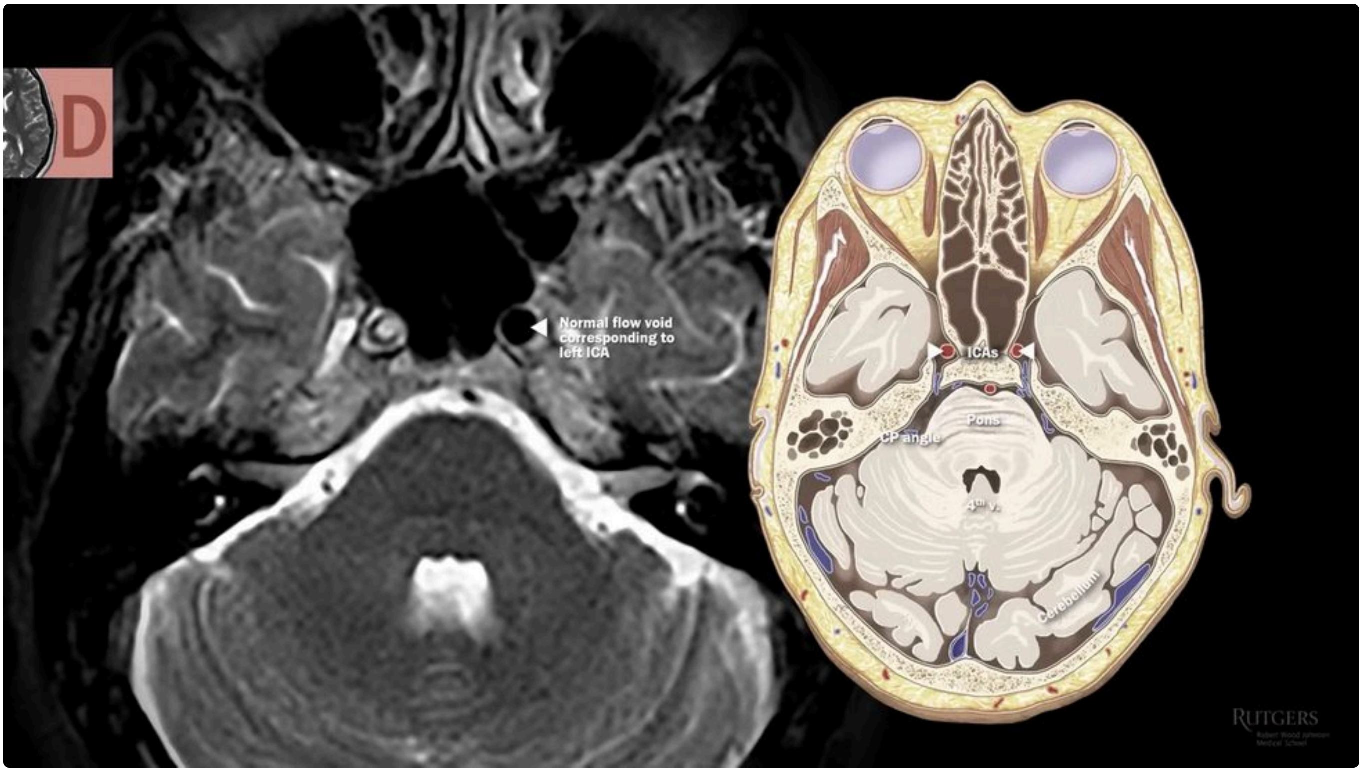
15 Dark On T2

Convenient symmetry: what is dark on T2 is essentially what is bright on T1.

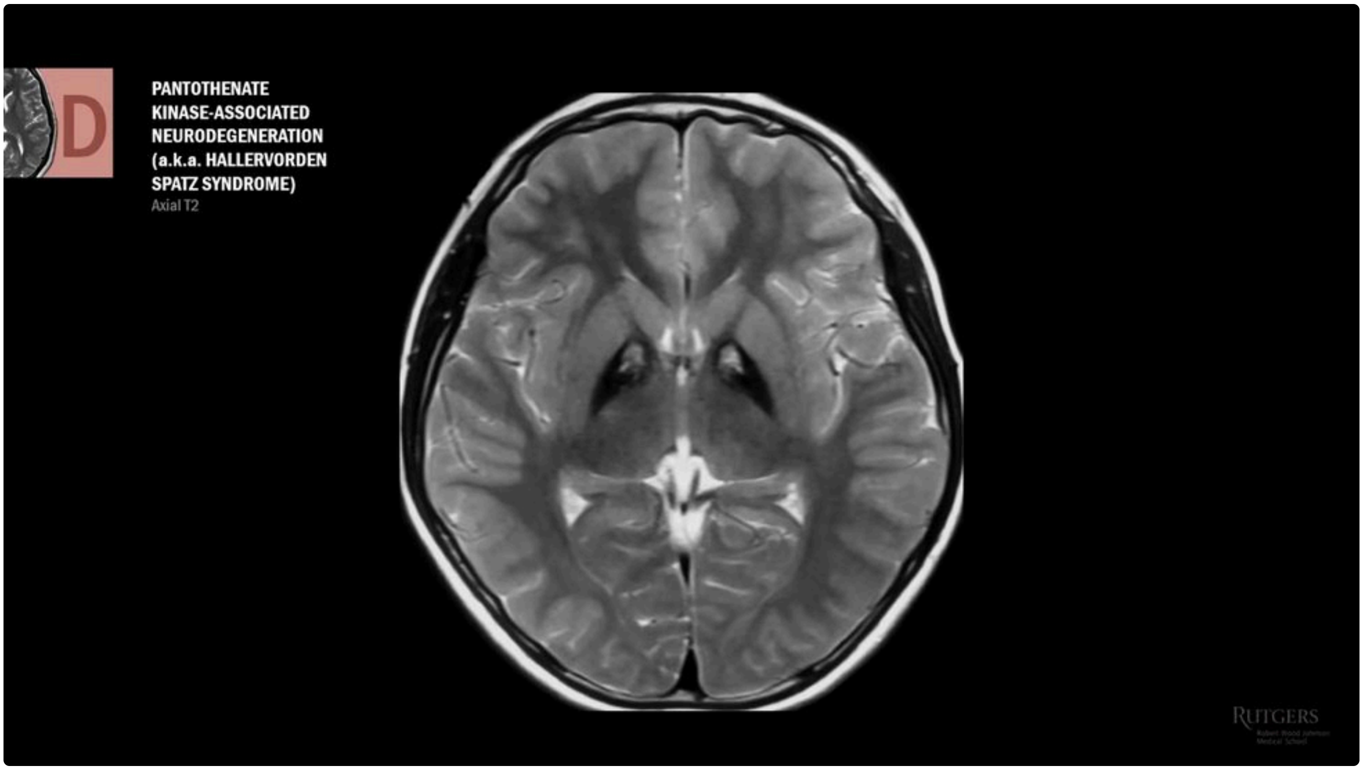
DARK ON T2	EXAMPLE
Protein-rich lesions	Colloid cyst
Paramagnetic substances	Iron, copper, melanin, calcium
Flowing blood	Normal flow voids
Acute haematoma	Within 3 days



Flowing blood is not imaged, creating dark flow voids. Identify the circle of Willis: internal carotids, MCAs, PComms, PCAs, and the confluence of sinuses.



A flow void present on one side and absent on the other. The right carotid is occluded.

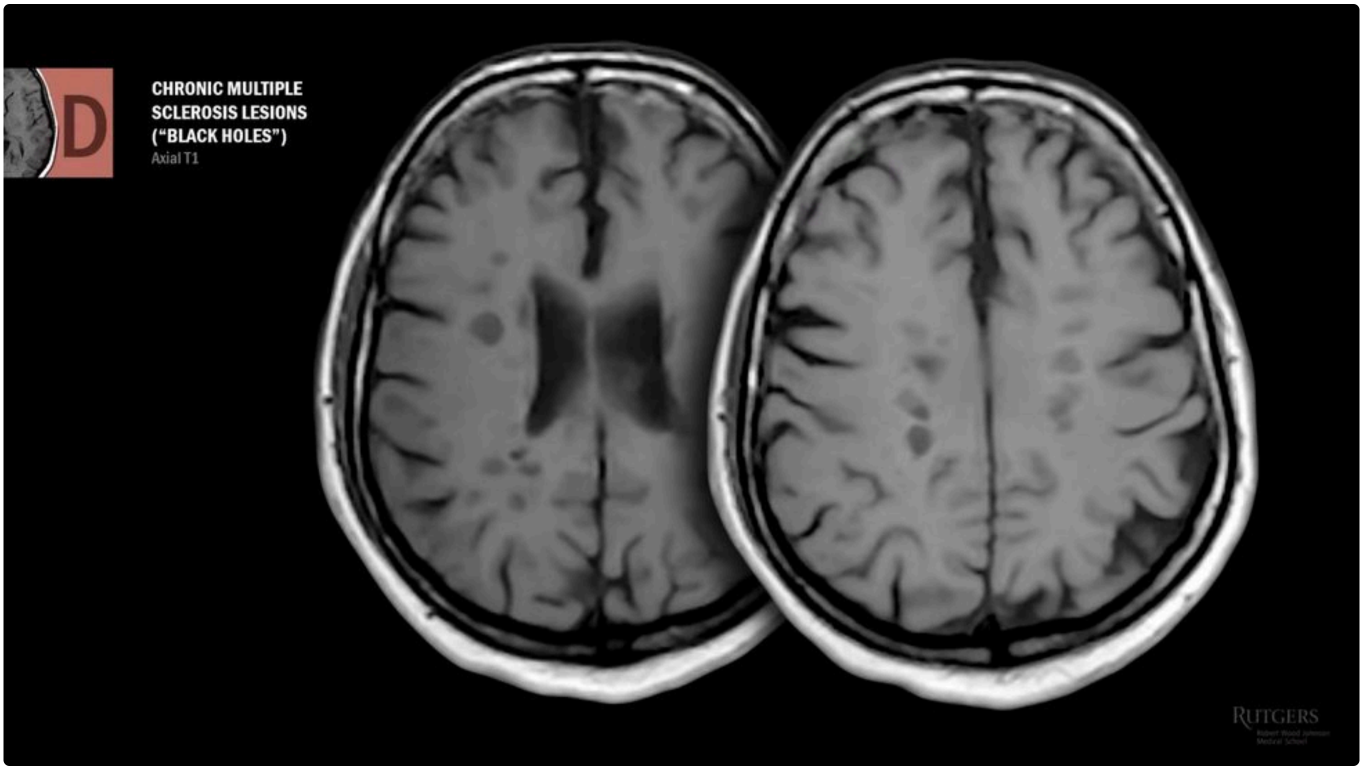


Dark iron accumulation with surrounding bright gliosis in the basal ganglia: the eye of the tiger sign, in pantothenate kinase-associated neurodegeneration.

> **REASON IT OUT, DO NOT MEMORISE IT**
> Wilson's copper was bright on T1. T2 is the opposite of T1. So iron and copper deposition is **dark on T2**, and you never had to learn that separately.

16 Dark On T1

T1 hypointensity is the simplest rule in the guide: **chronic lesions and fluid**. Chronic lesions are filled with fluid, so it is one rule, not two. In this respect T1 behaves like a non-contrast CT.

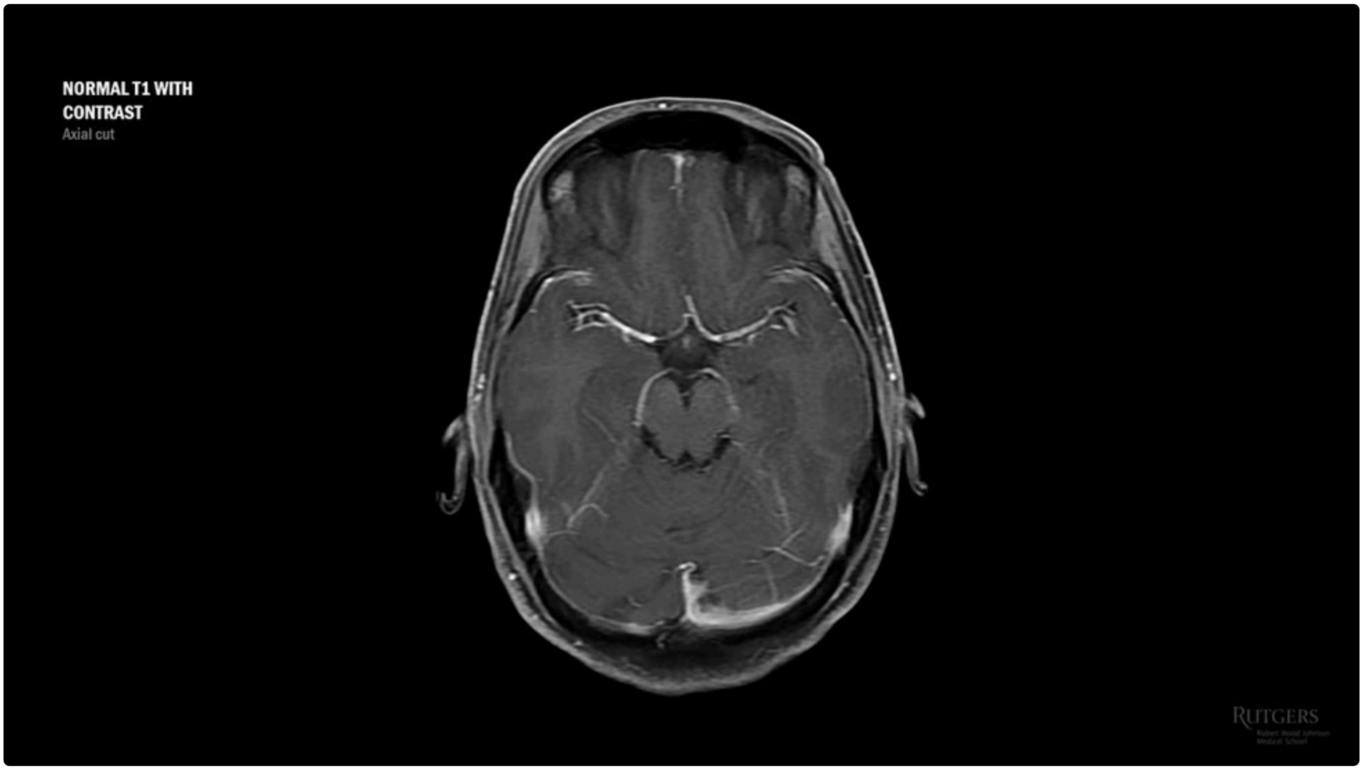


Chronic multiple sclerosis lesions on T1: black holes. Their number correlates with disability.

> **WHAT T1 IS REALLY FOR**
>
> T1 without contrast is generally the **least** diagnostic sequence. Its job is to be the baseline template against which every other sequence is compared.

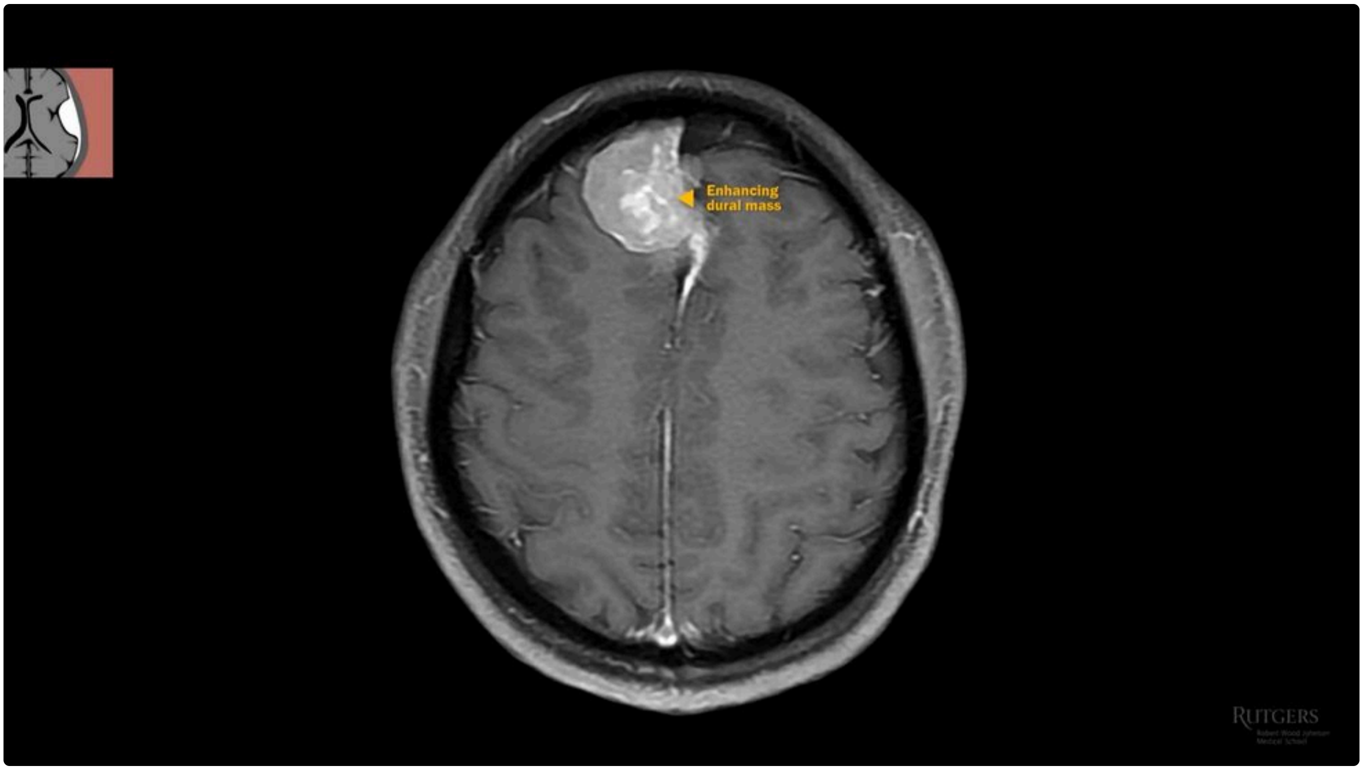
17 Enhancement Patterns

Gadolinium is bright on T1 and does not cross an intact blood brain barrier. Contrast in the sulci can be normal. **Contrast in the parenchyma or meninges never is.**



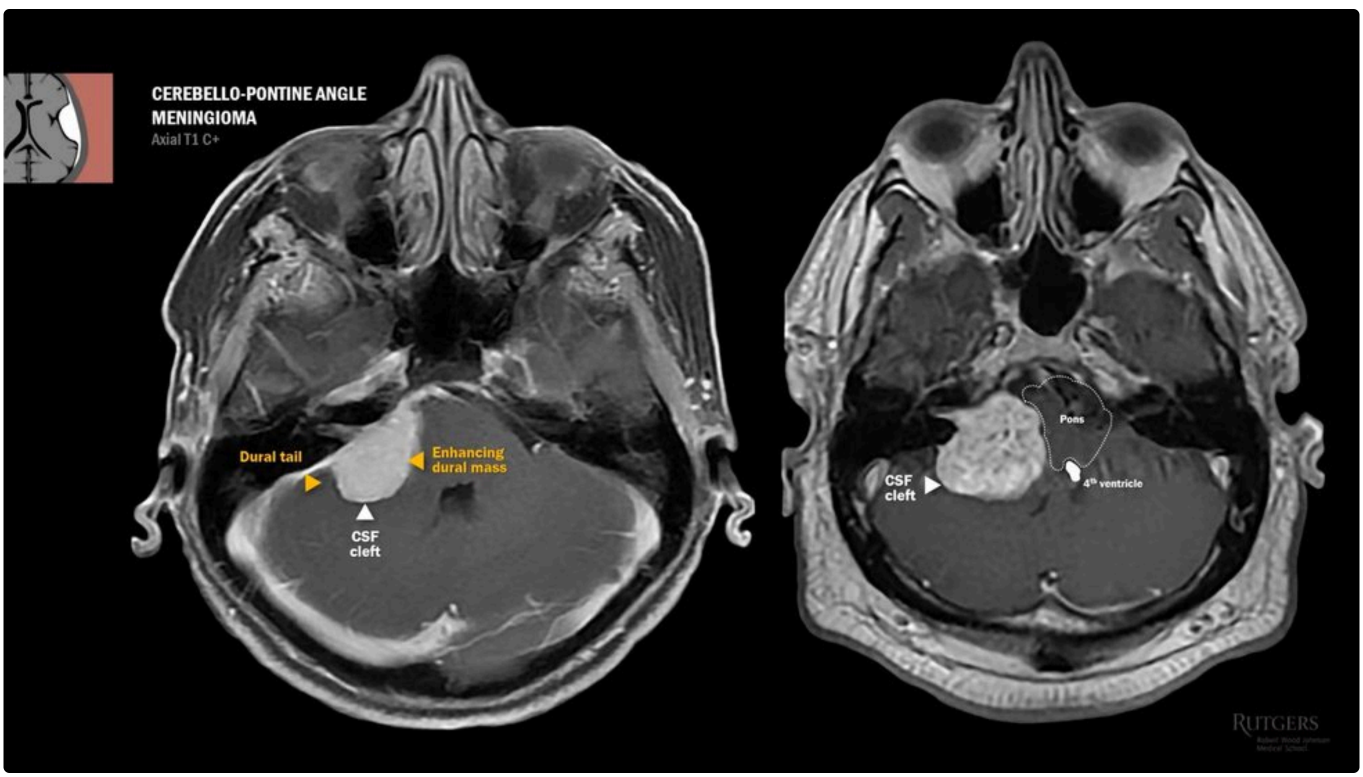
Normal T1 with contrast. Arteries, deep draining veins, transverse sinus and superior sagittal sinus. Fat has been subtracted so it is not mistaken for contrast.

PATTERN	COMPARTMENT	THINK OF
Dural tail	Extra-axial	Meningioma, dural metastasis
Leptomeningeal	Extra-axial	Bacterial, tuberculous or carcinomatous meningitis
Periventricular	Extra-axial	CNS lymphoma, subependymal spread
Ring	Intra-axial	Abscess, toxoplasmosis, lymphoma, glioblastoma
Subcortical nodular	Intra-axial	Metastasis, acute MS lesions
Mural nodule	Intra-axial	Pilocytic astrocytoma

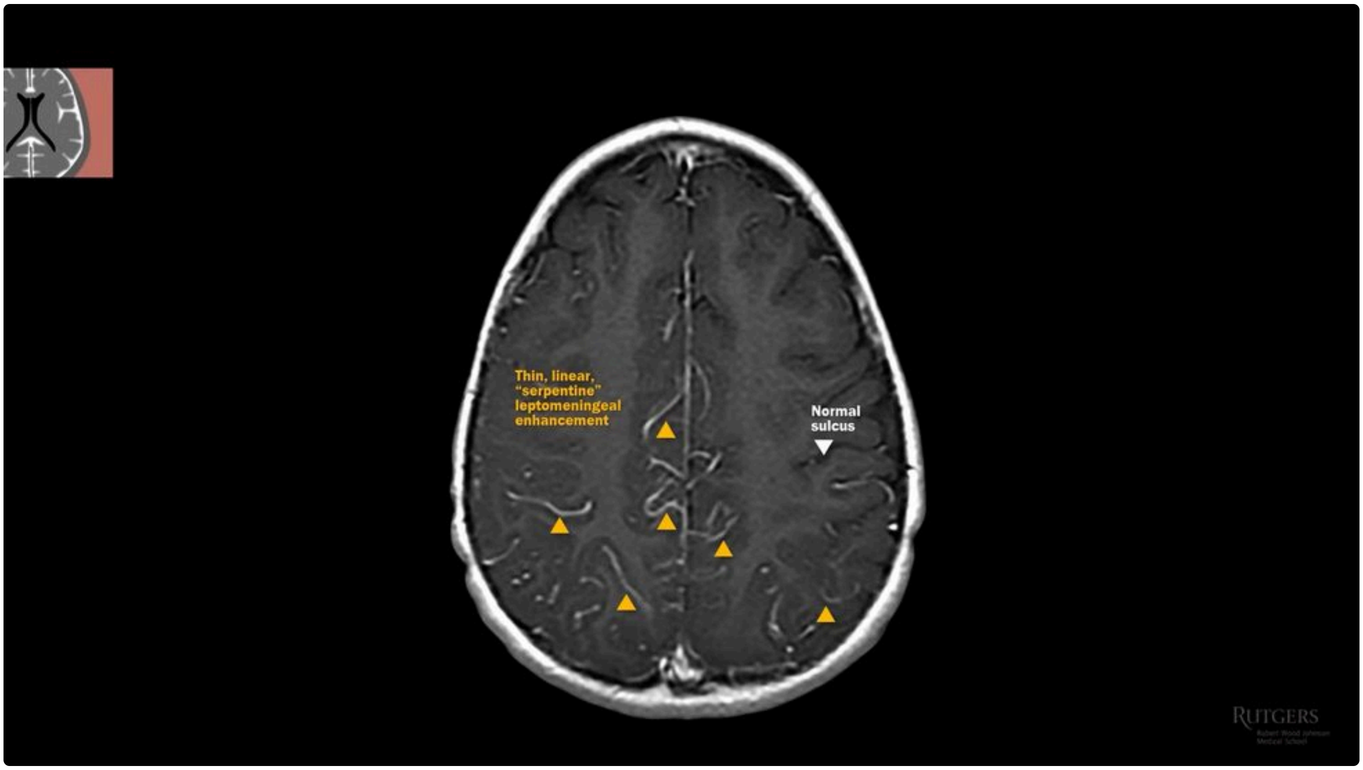


An enhancing dural mass. A dark CSF cleft separates it from parenchyma, proving extra-axial, and there is the dural tail. Meningioma.

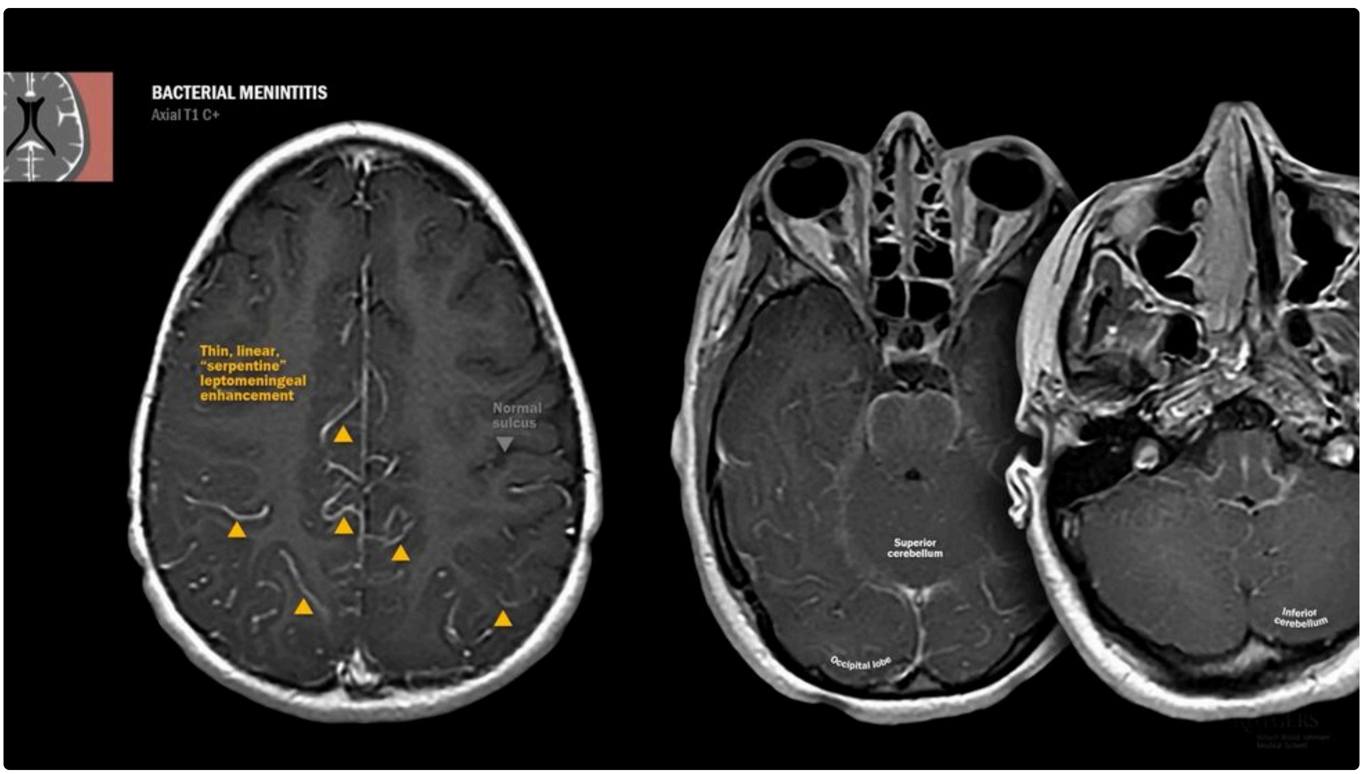
> **THE CSF CLEFT**
> A dark rim of CSF between mass and parenchyma is what tells you a lesion is **extra-axial**. The dural tail is largely reactive change, and it is **not unique to meningioma** - dural metastases have tails too.



Also at the cerebellopontine angle, but heterogeneous, with a head rather than a tail, extending into the internal acoustic canal. Vestibular schwannoma.



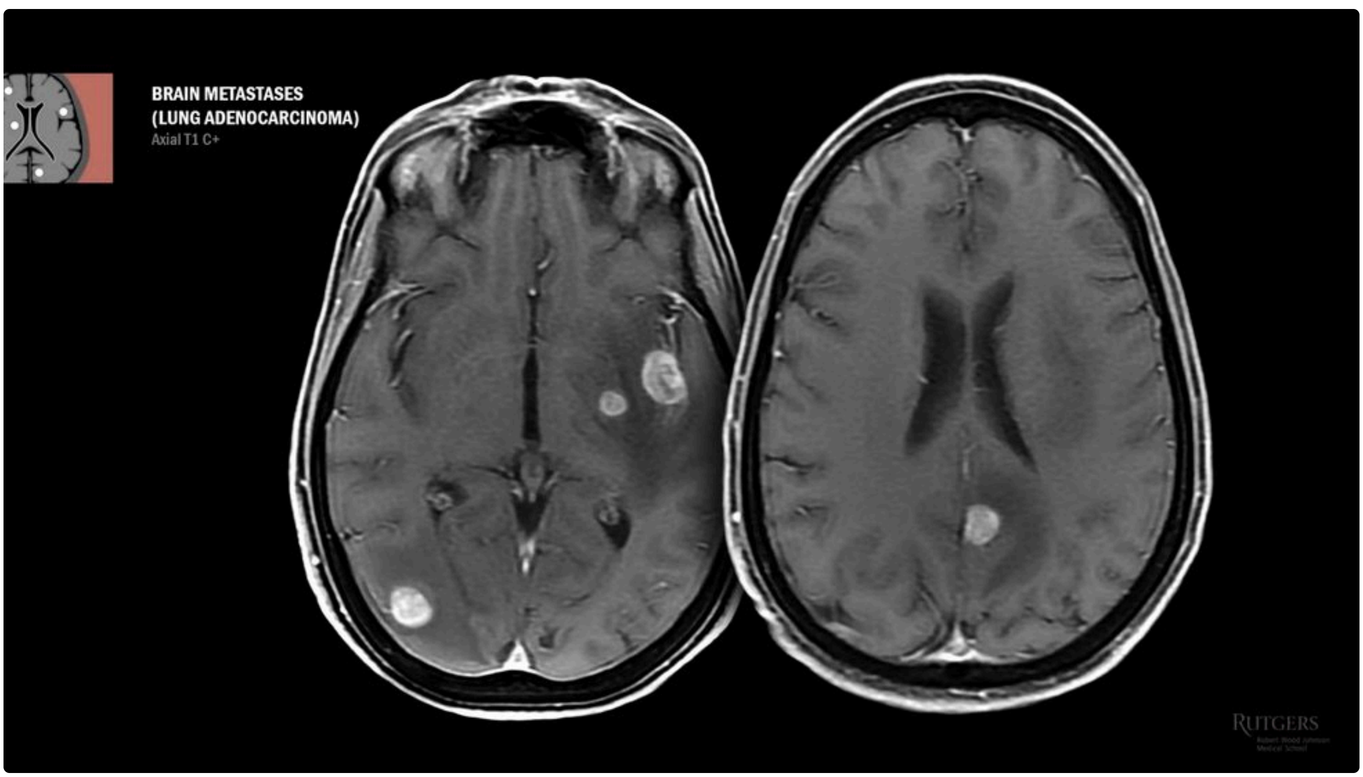
Normal sulci are dark on T1. Here there is leptomeningeal enhancement throughout. With fever and nuchal rigidity, bacterial meningitis.



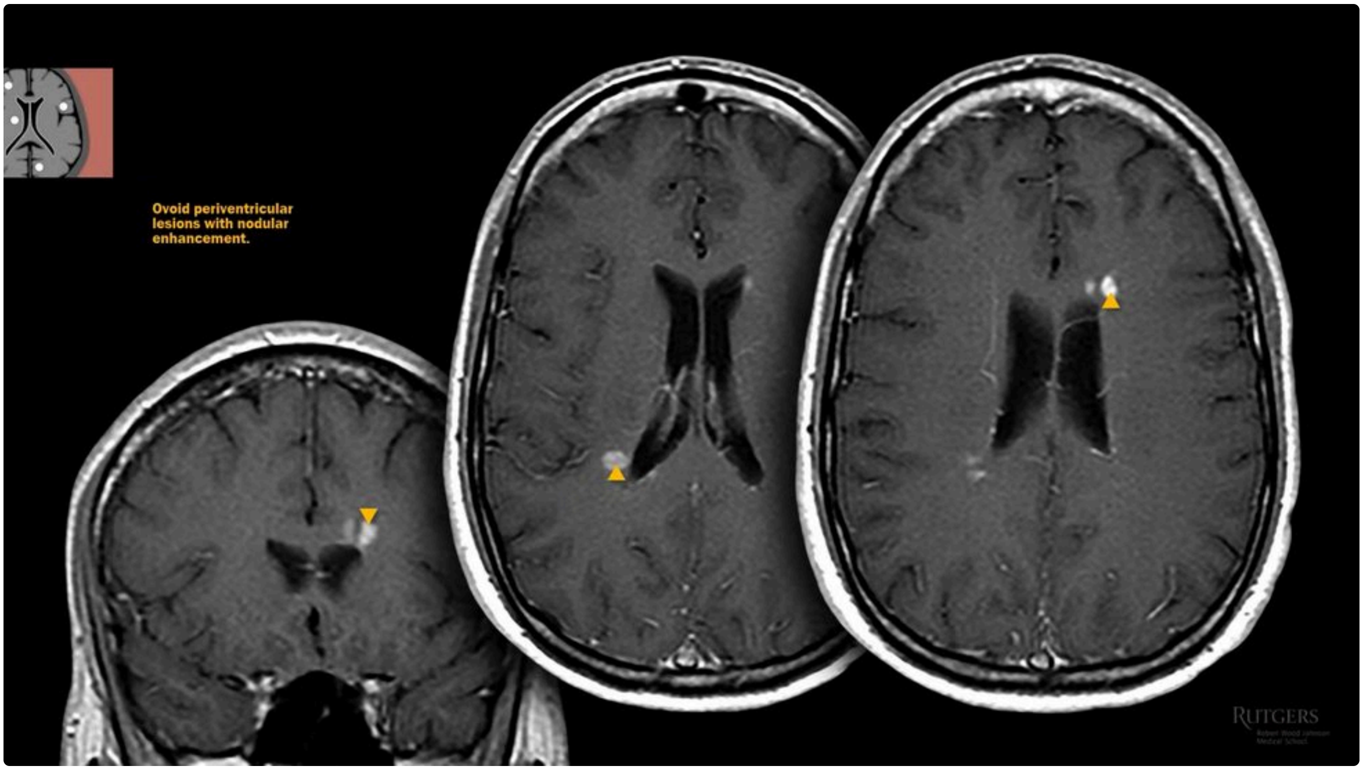
Subtler: enhancement around pons and medulla. Basilar meningitis from tuberculosis, the commonest CNS presentation of TB.

Leptomeningeal invasion by metastasis: carcinomatous meningitis, usually breast or lung.

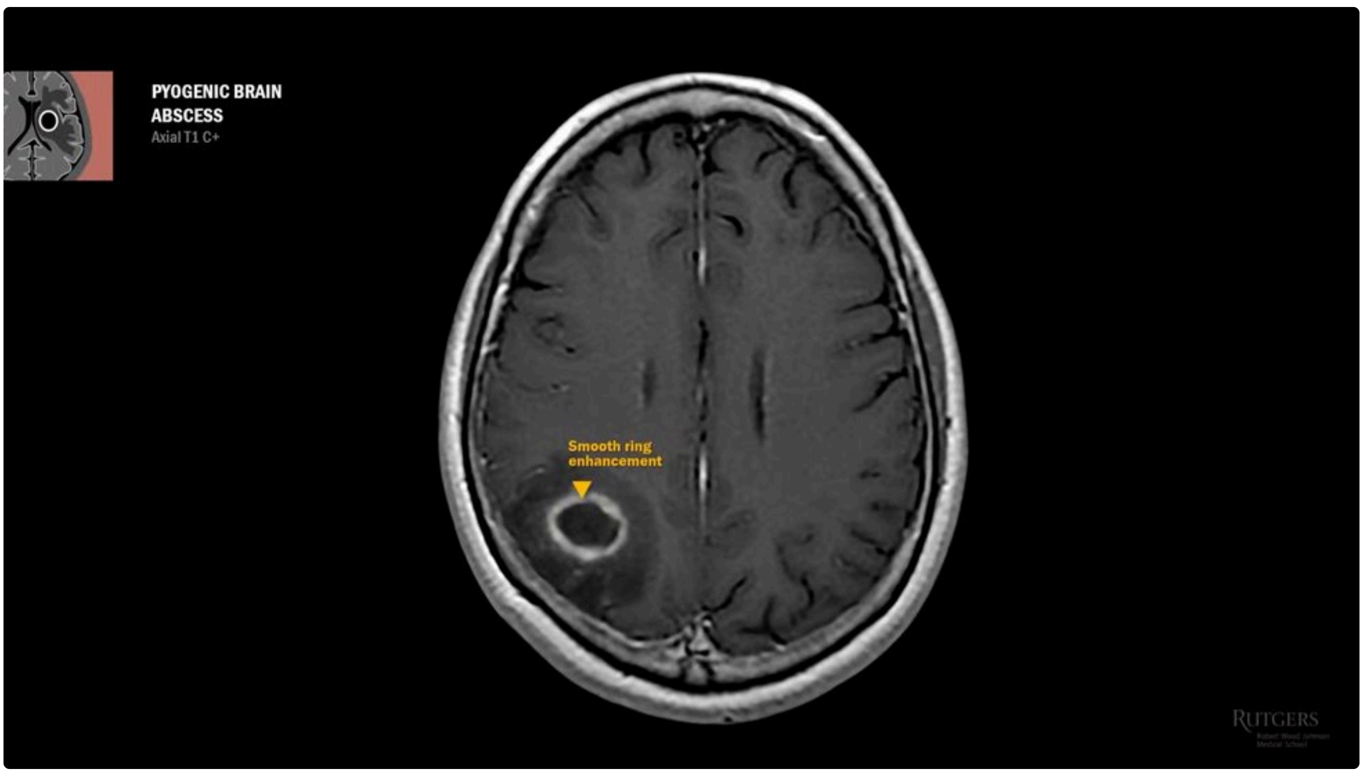
> **IMAGING CANNOT SEPARATE THESE THREE**
> Bacterial, tuberculous and carcinomatous meningitis look strikingly similar. **History and CSF sampling** differentiate them, not the scan.



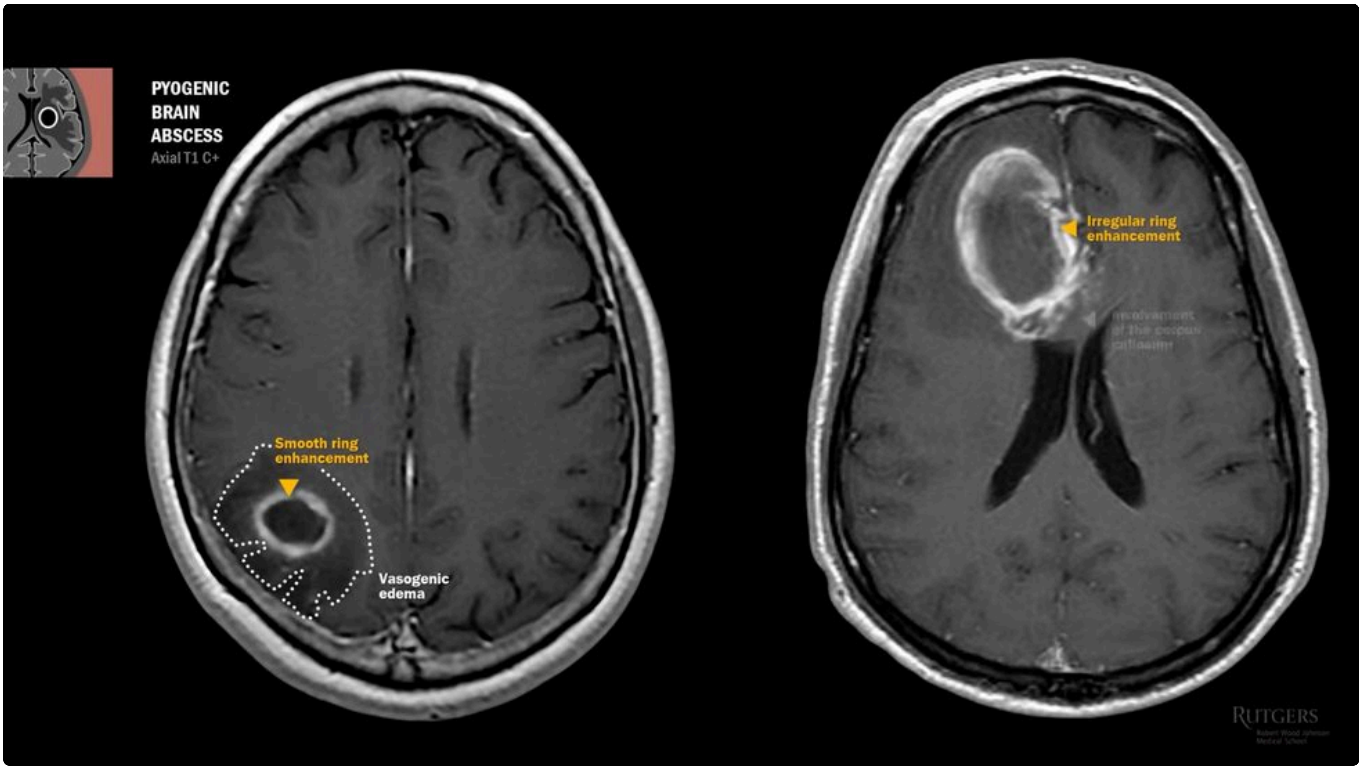
Multiple well-circumscribed enhancing masses at the grey-white junction with surrounding vasogenic oedema. Metastatic disease.



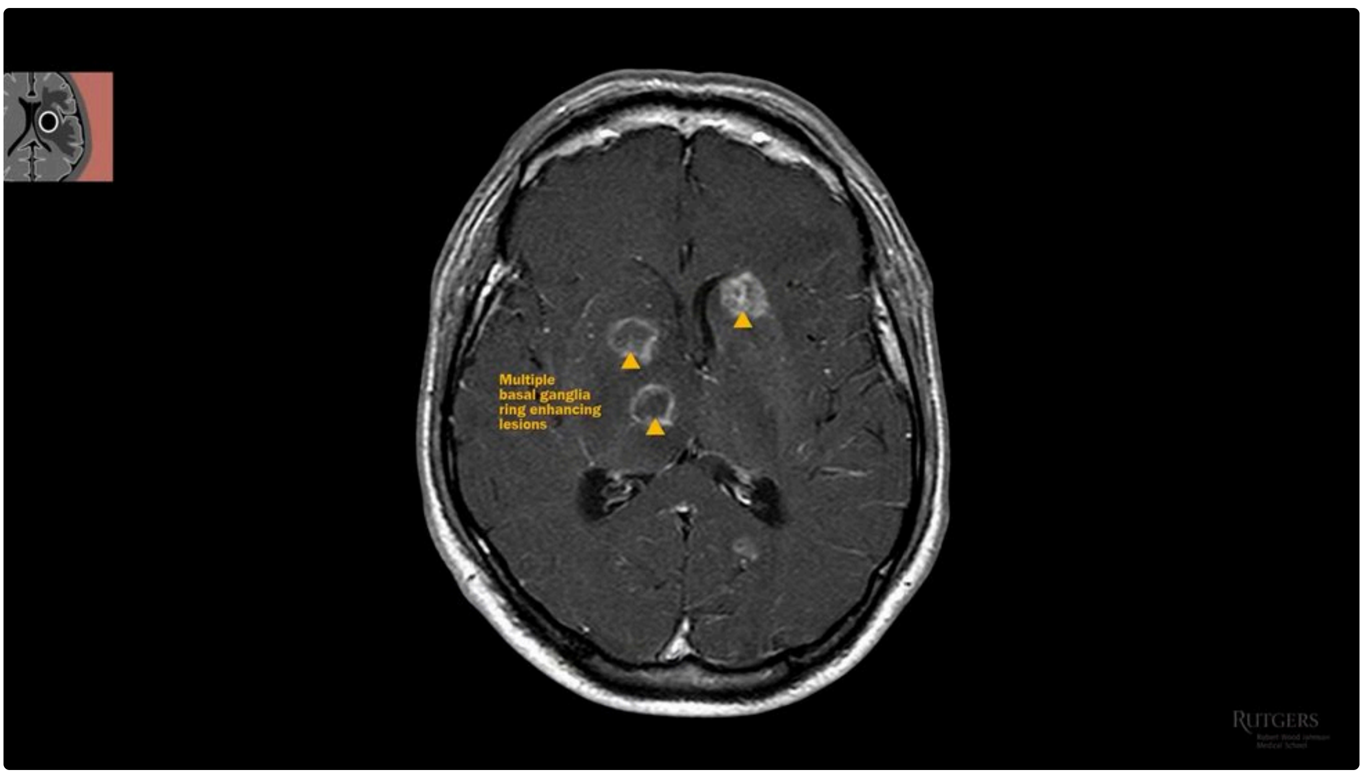
Subcortical enhancing nodules, but no oedema, and suspiciously ovoid and perpendicular. Acute multiple sclerosis lesions.



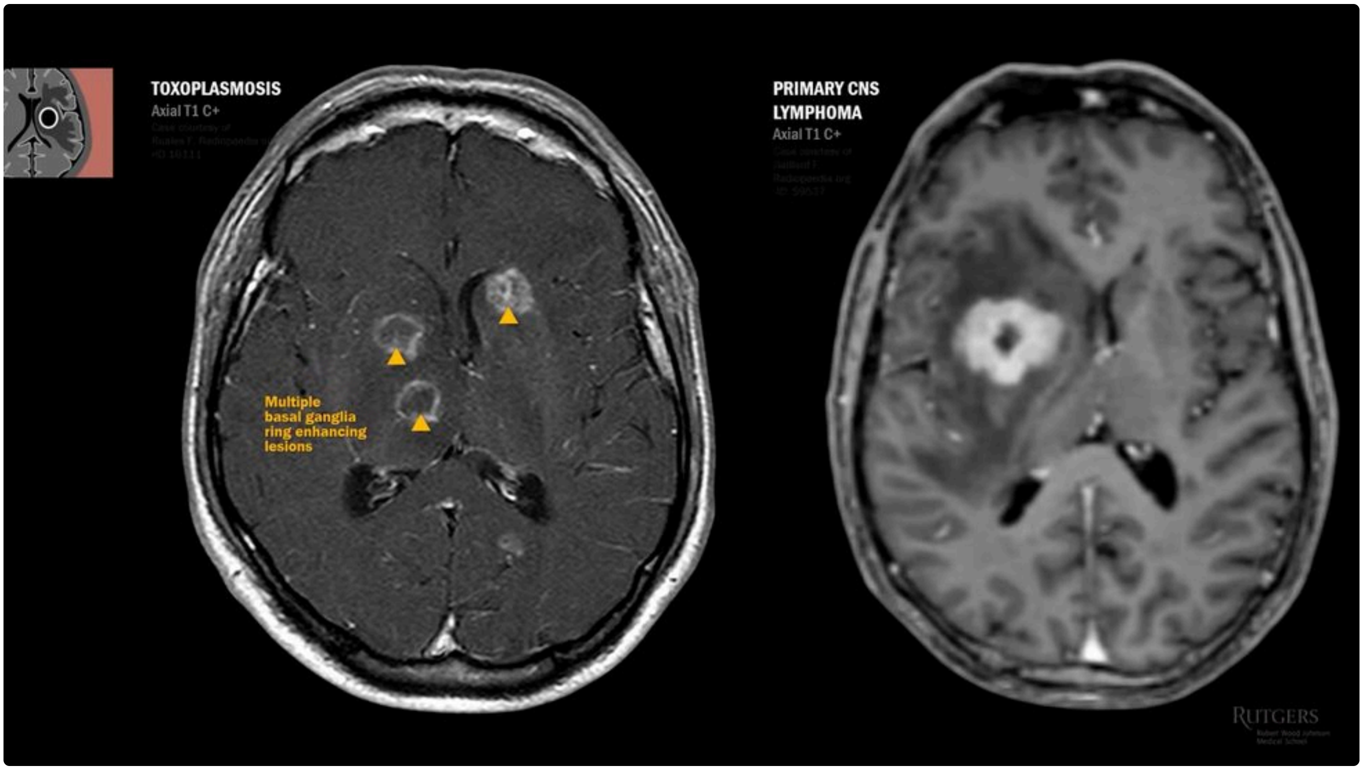
Smooth ring-enhancing capsule, necrotic centre, finger-like vasogenic oedema dark on T1. Pyogenic brain abscess.



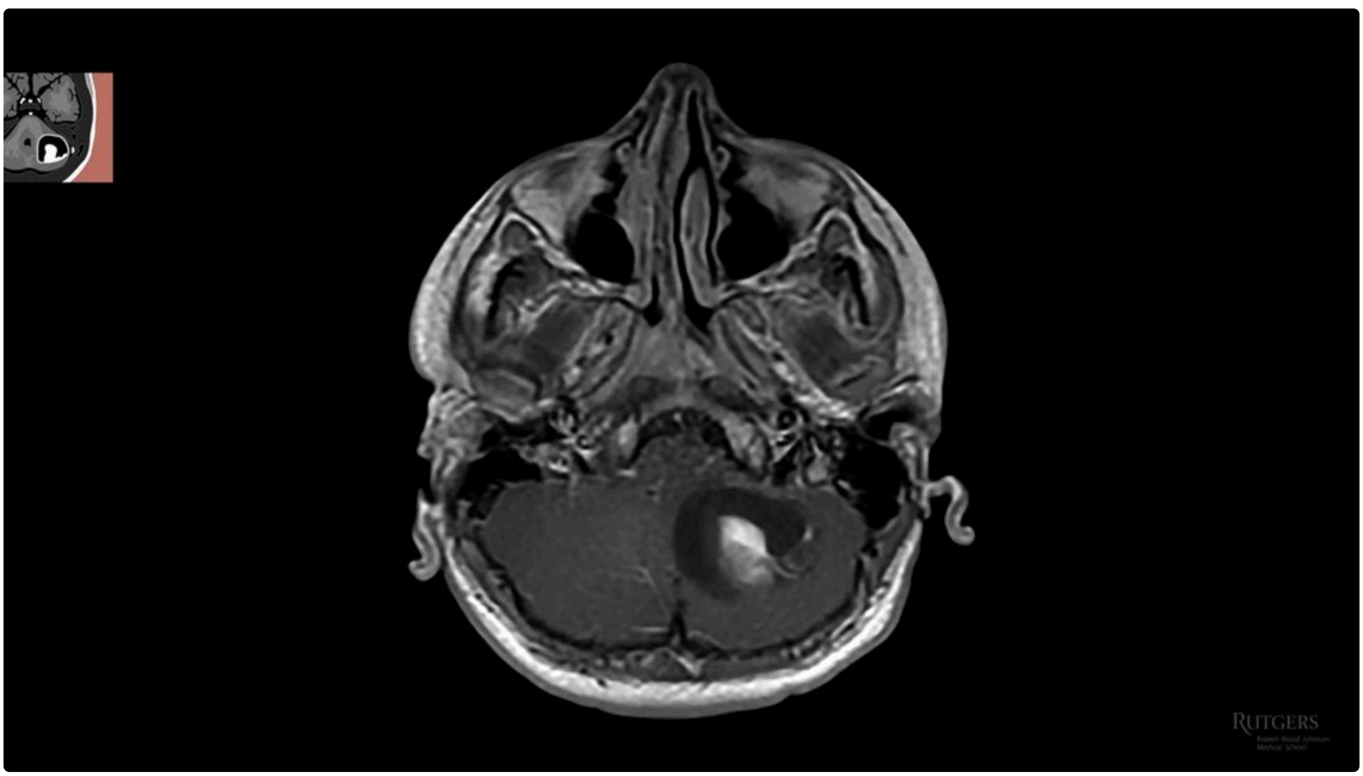
Also ring enhancing, but irregular, with corpus callosum involvement. High-grade glioma.



Multiple ring-enhancing lesions favouring basal ganglia in an HIV-positive patient: toxoplasmosis.



A large solitary ring-enhancing mass, and enhancement coating the ventricles. Periventricular and subependymal spread points to CNS lymphoma, not toxoplasmosis.

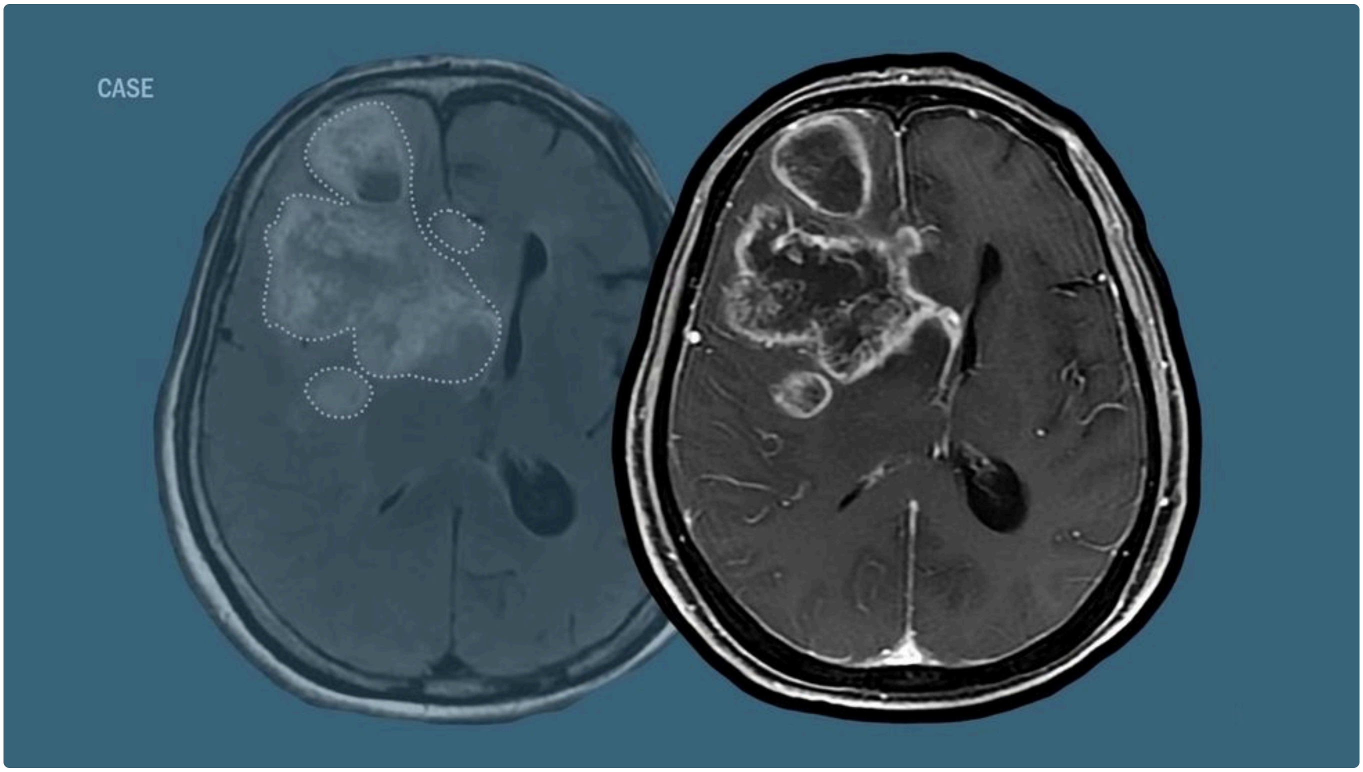


A cystic lesion with an enhancing mural nodule in a 14-year-old with truncal ataxia: pilocytic astrocytoma, the commonest infratentorial paediatric neoplasm.

The Case Solved

Back to the 73-year-old woman. Run the five steps.

STEP	FINDING
CT	Large confluent white matter hypodensity, plus a small hyperdensity
T1	Dark. Adds nothing over CT, so it is usually skipped
T2 / FLAIR	Huge hyperintensity. Finger-like subcortical projections sparing cortex, so **vasogenic**
The small dark focus	Dark on T2 means protein, flow void, paramagnetic substance or **acute blood**
T1 with contrast	**Irregular ring enhancement with satellite lesions, crossing the corpus callosum**



The contrast-enhanced study. Irregular ring enhancement, satellite lesions, corpus callosum crossed.

THE ANSWER

A mass with tremendous vasogenic oedema, a small acute haemorrhagic focus, and irregular ring enhancement crossing the corpus callosum. **This is a high-grade glioma.**

19 Quick Reference

Everything above, compressed to one screen.

SEQUENCE	BRIGHT MEANS	DARK MEANS
CT	Calcium, bone, acute blood	Fluid, chronic lesions, oedema
T2 / FLAIR	Oedema, hydrocephalus, inflammation, subacute blood	Iron and copper, protein, flow voids, acute blood
T1	Iron and copper, melanin, fat, protein, subacute blood	Chronic lesions and fluid
DWI + ADC	Bright DWI with **dark** ADC is true ischaemia or abscess	Bright DWI with bright ADC is T2 shine-through

THE MNEMONICS, IF YOU WANT THEM

S2HI2NE, bright on T2: Stroke and Subacute haematoma, Hydrocephalus, Ischaemic and Inflammatory lesions, Neoplasm, Edema.
SHIMMeR, bright on T1: Subacute haematoma, Iron, Metals, Melanoma, Rich in protein.
FAB PIC, dark on T1 and T2: Flowing or Acute Blood, Protein, Iron and paramagnetics, Chronic lesions.
MR Lights the Damage Very Nicely, enhancement: Mural nodule, Ring, Leptomeningeal, Dural tail, periVentricular, Nodular subcortical.

Dr. Rybinnik's closing instruction is the one that actually builds the skill: **read the study, commit to an impression, then compare it against the radiologist's report.** That is how you learn.

THE LINE TO KEEP

Imaging does not give you the diagnosis. **The history does.** What imaging gives you is a defensible differential, and only if you read it in the same order every time: identify, then symmetry, then density, then enhancement, then location.

SOURCES & NOTE

Sources: Brain Imaging, Crash Course - Dr. Rybinnik, The Neurophile, Rutgers Robert Wood Johnson Medical School Neurology, 58m06s, published March 29th 2020 (youtube.com/watch?v=DdPRfPm9SI4); All images are frames from that lecture and remain the property of Rutgers RWJMS; Anatomical illustrations shown within those frames are credited on screen to Netter.

A study aid built from a single lecture, for personal educational use, not a clinical reference and not for distribution. The lecture is rated MS for medical students by its author. Always read the study against the radiologist's report.

Glossary

Encephalomalacia Loss of brain tissue after injury, leaving a well-defined CSF-density defect.

Cytotoxic edema Cell-body swelling from energy failure, barrier intact. Grey matter involved.

Vasogenic edema Plasma leak through failed tight junctions. Finger-like, spares grey matter.

Neurovascular unit Astrocyte foot processes around capillary endothelium joined by tight junctions. The blood brain barrier.

FLAIR Fluid-Attenuated Inversion Recovery T2 with CSF signal subtracted, so oedema stands out.

DWI Diffusion Weighted Imaging Detects restricted diffusion of ischaemia within about 30 minutes.

ADC Apparent Diffusion Coefficient DWI's counterpart. True ischaemia is bright on DWI and dark here.

T2 shine-through A T2 or FLAIR bright lesion appearing bright on DWI without true restriction.

GRE Gradient Recalled Echo MRI sequence used to detect haemorrhage.

Extra-axial Outside the brain parenchyma. A CSF cleft is the giveaway.

Intra-axial Within the brain parenchyma.

Dense vessel sign A vessel visibly brighter than brain on CT, indicating acute clot.

Dural tail Reactive dural thickening beside an extra-axial mass. Suggestive of meningioma, not exclusive to it.

CSF cleft Dark rim of CSF separating a mass from parenchyma, proving it is extra-axial.

Transependymal flow CSF forced across ventricular walls under pressure. A sign of obstructive hydrocephalus.

Dawson's fingers Ovoid periventricular lesions of multiple sclerosis, aligned along central veins.

Hemosiderin Iron residue left by old haemorrhage. Dark rim on T2.

Hand knob Omega-shaped bump in the pre-central gyrus where hand motor control sits.

Uncus Medial temporal lobe lateral to the midbrain. The structure that herniates in uncal herniation.

Encephalomalacia vs cyst Encephalomalacia is irregular tissue loss; an arachnoid cyst is smooth and extra-axial.